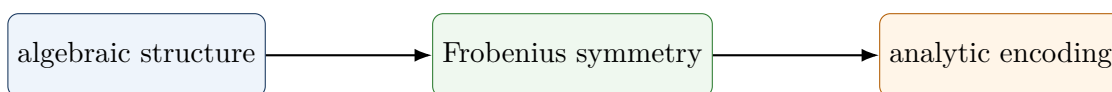


Mathematical Reboot for Chebotarev

Deep Intuition Edition

A self-contained refresher in proof, algebra, Galois theory, real analysis, and complex analysis



Revised and expanded for conceptual reconstruction

August 2026

Preface: what changed in this edition

This book is a reconstruction manual for mathematics that may once have been fluent and is now partly rusty. Its scope follows the mathematical preparation needed before a serious study of the Chebotarev density theorem: proof language; groups, actions, rings, and quotients; fields, polynomials, and Galois theory; limits, series, and uniform convergence; complex analysis; and the first bridge from Frobenius to Euler products.

The governing principle is that a definition is not truly remembered until you can answer four questions about it:

1. What problem forced mathematicians to invent this definition?
2. What mental picture makes the formal statement feel inevitable rather than arbitrary?
3. Which concrete examples separate the definition from nearby but weaker ideas?
4. Which proofs become natural once the definition is viewed correctly?

This edition therefore adds substantially more *connective tissue*. Every major concept is approached through motivation, an intuitive model, a formal statement, an example ladder, a hypothesis audit, and a forward connection. Proofs are not merely supplied; their architecture is explained before the details. The goal is to help you regenerate the mathematics later, not to make the pages easy to recognize now.

Deep intuition

A useful way to organize mathematical memory is *compression with regeneration*. A theorem is not a paragraph to memorize. It is a small machine: hypotheses enter, one decisive mechanism acts, and the conclusion comes out. If you retain the mechanism and why the hypotheses feed it, most of the details can be reconstructed.

Correctness and scope

The statements and examples in this revision were re-audited for hypotheses, edge cases, and hidden dependencies. In particular, the treatment of quotient well-definedness, Sylow theory, the finite Galois correspondence, finite-field Frobenius, differentiation under uniform convergence, and the topological hypotheses behind Cauchy's theorem has been rewritten to make the logical dependencies explicit.

We take the completeness axiom for $\mathbb{R}$ (equivalently, the least-upper-bound property) and standard

finite-dimensional linear algebra as foundational background. Everything else used as a named theorem in the main development is proved. When a topological definition is tailored to the analytic proof (as in the piecewise-smooth version of simple connectivity), that choice is stated explicitly rather than silently invoking a stronger theorem from topology.

How to use this as a refresher

For each section, use a four-pass loop:

1. **Predict.** Before reading the formal definition, ask what problem it should solve.
2. **Formalize.** Write the definition from memory, including quantifiers and hypotheses.
3. **Stress-test.** Produce an example, a nonexample, and a boundary case.
4. **Reconstruct.** Close the book and reproduce the proof architecture before filling in details.

Recognition is weak evidence. A more reliable mastery test is: can you explain the idea to someone else, manufacture a new example, and recover the decisive proof step without seeing the page?

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Part I

Proof language: recovering the grammar

Logic, definitions, and proof reconstruction

1.1 Statements, predicates, and quantifiers

Why this idea exists

Mathematical proofs fail surprisingly often not because the algebra is difficult, but because the logical shape of the statement has been misread. Quantifiers tell you who gets to choose what, and in what order. That order is part of the mathematics.

Definition 1.1. A *statement* is a sentence that is either true or false. A *predicate* $P(x)$ is a statement depending on a variable. The universal and existential quantifiers are

$$\forall x \in X P(x), \quad \exists x \in X P(x).$$

The first says that every allowed input passes the test; the second says that at least one does.

Deep intuition

Think of quantifiers as a game of choices. In

$$\forall x \exists y P(x, y),$$

you see x first and may choose y in response. In

$$\exists y \forall x P(x, y),$$

you must choose one y in advance that works for every x . The second demand is usually much stronger.

Example ladder

Easy. $\forall n \in \mathbb{Z} \exists m \in \mathbb{Z}$ such that $m > n$ is true: choose $m = n + 1$.

Stronger order. $\exists m \in \mathbb{Z} \forall n \in \mathbb{Z}$ such that $m > n$ is false. One integer cannot dominate

all integers.

Analysis preview. Sequence convergence says

$$\forall \varepsilon > 0 \exists N \forall n \geq N : |a_n - L| < \varepsilon.$$

The index N may depend on the requested accuracy ε . Uniform convergence will add a variable x but require the same N to work for all x .

Negation reverses the quantifier and negates the predicate:

$$\neg(\forall x P(x)) \iff \exists x \neg P(x), \quad \neg(\exists x P(x)) \iff \forall x \neg P(x).$$

Repeatedly applying this rule gives, for example,

$$\neg(\forall x \exists y P(x, y)) \iff \exists x \forall y \neg P(x, y).$$

Example 1.2 (Negating convergence). The assertion $a_n \rightarrow L$ means

$$\forall \varepsilon > 0 \exists N \forall n \geq N : |a_n - L| < \varepsilon.$$

Its negation is

$$\exists \varepsilon_0 > 0 \forall N \exists n \geq N : |a_n - L| \geq \varepsilon_0.$$

The intuitive meaning is important: failure to converge to L means there is one fixed tolerance ε_0 that is violated arbitrarily far out. It is not enough that the sequence occasionally misses a tolerance that itself keeps shrinking.

Common pitfall

Do not negate $|a_n - L| < \varepsilon$ as $|a_n - L| > \varepsilon$. The correct negation is $|a_n - L| \geq \varepsilon$. Boundary symbols matter.

1.2 Implication, converse, and contrapositive

Definition 1.3. The implication $P \Rightarrow Q$ says that whenever P is true, Q must be true. We call P a *sufficient* condition for Q , and Q a *necessary* condition for P . The *converse* is $Q \Rightarrow P$. The *contrapositive* is $\neg Q \Rightarrow \neg P$.

Deep intuition

A sufficient condition is a trigger: once P happens, Q is forced. A necessary condition is a checkpoint: if P happened, Q must already have been present. In a diagram,

$$P \implies Q.$$

The arrow points from sufficient to necessary.

Proposition 1.4. An implication is logically equivalent to its contrapositive.

Proof. The implication $P \Rightarrow Q$ is false in exactly one situation: P is true and Q is false. The contrapositive $\neg Q \Rightarrow \neg P$ is false in exactly one situation: $\neg Q$ is true and $\neg P$ is false, which is the same situation. Therefore they always have the same truth value. $\square$

Example ladder

Let P be “ $12 \mid n$ ” and Q be “ $6 \mid n$.” Then $P \Rightarrow Q$. Divisibility by 12 is sufficient for divisibility by 6, while divisibility by 6 is necessary for divisibility by 12. The converse is false: $n = 6$ is a counterexample.

For a geometric example, “a quadrilateral is a square” implies “it is a rectangle.” Being a rectangle is necessary but not sufficient for being a square.

1.3 Four proof patterns and how to choose among them

1.3.1 Direct proof

To prove $P \Rightarrow Q$ directly, assume P , unpack definitions, and transform the available information until Q becomes visible.

Example 1.5 (Even squares). If n is even, write $n = 2k$. Then $n^2 = 4k^2 = 2(2k^2)$, so n^2 is even. The decisive move is not the multiplication; it is translating the adjective “even” into the usable equation $n = 2k$.

1.3.2 Contrapositive

Use the contrapositive when the negation of the conclusion has a concrete structure.

Example 1.6. To prove “if n^2 is even, then n is even,” prove the contrapositive: if n is odd, write $n = 2k + 1$; then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

so n^2 is odd.

1.3.3 Contradiction

To prove P , assume $\neg P$ and derive an impossibility. Contradiction is especially natural for impossibility and infinitude statements, where the negation gives a finite or rigid object to attack.

1.3.4 Induction

To prove $P(n)$ for all $n \geq n_0$, establish $P(n_0)$, then prove $P(k) \Rightarrow P(k + 1)$ for arbitrary $k \geq n_0$.

Deep intuition

Induction is a propagation theorem. The base case lights the first lamp; the induction step proves that any lit lamp forces the next one to light. The induction hypothesis is not assuming the theorem globally. It temporarily assumes one instance in order to establish the next.

Example 1.7 (Arithmetic series). We prove $1 + \cdots + n = n(n+1)/2$. For $n = 1$ it is true. Assume

$$1 + \cdots + k = \frac{k(k+1)}{2}.$$

Then

$$1 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Thus the formula holds for $k+1$, completing the induction.

1.4 Equality, existence, and uniqueness

To prove equality of sets $A = B$, prove $A \subseteq B$ and $B \subseteq A$. To prove functions $f, g : X \rightarrow Y$ are equal, prove $f(x) = g(x)$ for every $x \in X$. These are not stylistic conventions; they simply unpack what equality means for the objects involved.

A statement “there exists a unique x such that $P(x)$ ” contains two logically separate claims:

1. *existence*: at least one such x exists;
2. *uniqueness*: any two such objects must coincide.

Often the two parts use different ideas.

Example 1.8 (Uniqueness of identity). Suppose e and e' are both identity elements in a group. Then $ee' = e'$ because e is an identity, while $ee' = e$ because e' is an identity. Hence $e = e'$.

Deep intuition

A common uniqueness trick is to make two hypothetical candidates interact inside one expression. The defining property can then be applied in two different ways, forcing equality.

1.5 Definitions as compressed theorems

Why this idea exists

Graduate-level definitions often look like lists of clauses. The clauses are rarely arbitrary. Each one is usually the minimal repair needed to prevent some failure in a later construction.

For instance, if H is to be a subgroup of G , we need to be able to multiply and invert elements of H without leaving H . Yet these requirements can be compressed.

Proposition 1.9 (One-step subgroup test). *A nonempty subset $H \subseteq G$ is a subgroup if and only if*

$$a, b \in H \implies ab^{-1} \in H.$$

Proof. If H is a subgroup, the condition is immediate. Conversely, assume $H \neq \emptyset$ and the displayed condition. Choose $a \in H$. Taking $a = b$ gives $e = aa^{-1} \in H$. Taking $a = e$ gives $b^{-1} \in H$ for every $b \in H$. Finally, for $a, b \in H$, we have $b^{-1} \in H$, so applying the condition to a and b^{-1} gives

$$a(b^{-1})^{-1} = ab \in H.$$

Thus H contains the identity and is closed under inverses and products; associativity is inherited from G . $\square$

Deep intuition

The expression ab^{-1} means “do a , then undo b .” Closure under this one kind of comparison is strong enough to recover all subgroup structure.

1.6 Three model proofs worth owning

Theorem 1.10 (Euclid). *There are infinitely many prime numbers.*

Proof. Assume for contradiction that the complete list of primes is $p_1, \dots, p_r$. Form

$$N = p_1 p_2 \cdots p_r + 1.$$

Since $N > 1$, it has a prime divisor q . The assumed list is complete, so $q = p_i$ for some i . But p_i divides both $p_1 \cdots p_r$ and N , hence divides their difference 1, impossible. Therefore the list of primes cannot be finite. $\square$

Proof architecture

Input: assume a finite complete list. **Key move:** manufacture a new integer that leaves remainder 1 modulo every listed prime. **Output:** any prime factor of that integer must be missing from the list.

Theorem 1.11 (Polynomial root bound). *Let K be a field and let $0 \neq f \in K[x]$ have degree d . Then f has at most d roots in K .*

Proof. We argue by induction on d . A nonzero constant has no roots. Suppose $d \geq 1$ and a is a root of f . By polynomial division, $f = (x - a)g$ with $\deg g = d - 1$. Every root $b \neq a$ of f satisfies

$$0 = f(b) = (b - a)g(b).$$

Because K is a field and $b - a \neq 0$, we get $g(b) = 0$. Thus all roots except possibly a are roots of g , of which there are at most $d - 1$ by induction. Hence f has at most d roots. $\square$

Hypothesis audit

The field hypothesis is being used through the absence of zero divisors. The theorem remains true over any integral domain. It can fail over rings with zero divisors: in $\mathbb{Z}/8\mathbb{Z}$, the degree-two polynomial $x^2 - 1$ has four roots 1, 3, 5, 7.

Theorem 1.12 (Positive minimum on a compact interval). *If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous and $f(x) > 0$ for every x , then there exists $m > 0$ such that $f(x) \geq m$ for all $x \in [0, 1]$.*

Proof. By the extreme value theorem, a continuous function on the compact interval $[0, 1]$ attains its minimum: there exists x_0 with $f(x_0) \leq f(x)$ for all x . By hypothesis $f(x_0) > 0$. Taking $m = f(x_0)$ proves the claim. $\square$

Why this matters later

This proof pattern recurs constantly: compactness turns pointwise qualitative information into a uniform quantitative bound. Later, compactness of closed disks and intervals is what lets uniform estimates work in analysis.

Part II

Algebra I: structure, symmetry, and quotients

Groups, homomorphisms, quotients, and actions

2.1 Groups as reversible processes

Why this idea exists

A group is the correct abstraction when we care about operations that can be composed and undone. The motivating examples are not numbers but transformations: rotate an object, permute labels, change coordinates, or apply an automorphism.

Definition 2.1. A *group* is a set G with a binary operation $(g, h) \mapsto gh$ such that:

1. $(gh)k = g(hk)$ for all $g, h, k \in G$;
2. there is $e \in G$ with $eg = ge = g$ for every g ;
3. for every g there is g^{-1} with $gg^{-1} = g^{-1}g = e$.

If $gh = hg$ for all g, h , the group is *abelian*.

Deep intuition

A group is a language of reversible moves. Associativity says a sequence of moves has an unambiguous total effect; the identity is “do nothing”; the inverse is “undo.” Commutativity is not required because real transformations often depend on order: rotate a book and then flip it, versus flip it and then rotate it.

Example ladder

Additive arithmetic. $(\mathbb{Z}, +)$ is a group; the inverse of n is $-n$.

Rotations. The rotations of an equilateral triangle form a cyclic group C_3 .

All triangle symmetries. Including reflections gives a group of six elements, isomorphic to S_3 . This is nonabelian.

Matrices. $\text{GL}_n(\mathbb{R})$ is the group of invertible $n \times n$ matrices. Singular matrices are excluded precisely because they cannot be undone.

Near-miss. $(\mathbb{Z}, \cdot)$ is not a group: most integers do not have multiplicative inverses in $\mathbb{Z}$.

2.2 Subgroups: closed subsystems

A subgroup is a collection of allowed moves that remains self-contained under composition and reversal. The one-step subgroup test from Chapter 1 is often the quickest criterion.

Example 2.2. For $n \geq 0$, $n\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$. It is nonempty, and if $a = nr$, $b = ns$, then

$$a - b = n(r - s) \in n\mathbb{Z}.$$

The additive form of the one-step test is closure under $a - b$.

Example 2.3. The set of positive real numbers is a subgroup of $(\mathbb{R}^\times, \cdot)$, since $ab^{-1} > 0$ whenever $a, b > 0$. By contrast, the positive integers are not a subgroup of $(\mathbb{Q}^\times, \cdot)$ because inverses such as $1/2$ leave the set.

2.3 Homomorphisms: maps that preserve behavior

Definition 2.4. A map $\varphi : G \rightarrow H$ is a *group homomorphism* if

$$\varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$$

for all $g_1, g_2 \in G$. Its kernel and image are

$$\ker \varphi = \{g : \varphi(g) = e_H\}, \quad \text{im } \varphi = \{\varphi(g) : g \in G\}.$$

Deep intuition

A homomorphism is a change of resolution that preserves the algebra. It may forget distinctions, but it is not allowed to distort the operation. The kernel is exactly the information that becomes invisible; the image is what survives.

Example ladder

Reduction modulo n . $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $k \mapsto k \bmod n$, forgets multiples of n . Its kernel is $n\mathbb{Z}$.

Exponential wrapping. $\varphi : (\mathbb{R}, +) \rightarrow S^1$, $t \mapsto e^{2\pi i t}$, wraps the real line around the circle. Its kernel is $\mathbb{Z}$: real numbers differing by an integer land at the same point.

Parity of a permutation. $\text{sgn} : S_n \rightarrow \{\pm 1\}$ records only whether a permutation is even or odd. Its kernel is A_n .

Proposition 2.5. For a group homomorphism $\varphi : G \rightarrow H$:

1. $\varphi(e_G) = e_H$;
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$;

3. $\ker \varphi$ is a normal subgroup of G ;

4. $\operatorname{im} \varphi$ is a subgroup of H .

Proof. Since $\varphi(e) = \varphi(e^2) = \varphi(e)^2$, multiply by $\varphi(e)^{-1}$ to obtain $\varphi(e) = e$. Then

$$e = \varphi(e) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}),$$

so $\varphi(g^{-1}) = \varphi(g)^{-1}$. The image is a subgroup by the subgroup test. The kernel is also a subgroup; moreover if $k \in \ker \varphi$ and $g \in G$, then

$$\varphi(gkg^{-1}) = \varphi(g)e\varphi(g)^{-1} = e,$$

so $gkg^{-1} \in \ker \varphi$. Thus the kernel is normal. $\square$

2.4 Normal subgroups and quotient groups

Why this idea exists

A quotient group is what you get when you decide that certain differences no longer matter. If elements that differ by an element of N are to be treated as equivalent, the multiplication of equivalence classes must not depend on which representatives we happened to choose. Normality is exactly the condition that makes this possible.

Definition 2.6. A subgroup $N \leq G$ is *normal*, written $N \triangleleft G$, if

$$gNg^{-1} = N \quad \text{for every } g \in G.$$

Equivalently, $gN = Ng$ for every g .

The left cosets gN are the equivalence classes for

$$g \sim h \iff h^{-1}g \in N.$$

We want to define $(gN)(hN) = (gh)N$.

Theorem 2.7 (Quotient criterion). *The rule $(gN)(hN) = (gh)N$ is well-defined on the set of left cosets if and only if $N \triangleleft G$. In that case the cosets form a group G/N .*

Proof. Assume N normal. If $g' = gn_1$ and $h' = hn_2$, then

$$g'h' = gn_1hn_2 = gh(h^{-1}n_1h)n_2.$$

Normality gives $h^{-1}n_1h \in N$, so $g'h'$ and gh lie in the same coset. Thus multiplication is well-defined. The group axioms descend from G .

Conversely, suppose coset multiplication is well-defined. Since $gN = gnN$ for $n \in N$,

$$(gN)(g^{-1}N) = (gnN)(g^{-1}N).$$

Thus $N = gng^{-1}N$, so $gng^{-1} \in N$. Hence $gNg^{-1} \subseteq N$; replacing g by g^{-1} gives equality. $\square$

Deep intuition

A quotient does not “divide by a subgroup” in the numerical sense. It changes what counts as distinguishable. In $\mathbb{Z}/n\mathbb{Z}$, integers that differ by a multiple of n become the same state. In $\mathbb{R}/\mathbb{Z}$, points one unit apart become identical, turning a line into a circle. In G/N , multiplying by an element of N becomes invisible.

Example ladder

Integers modulo n . $\mathbb{Z}/n\mathbb{Z}$ is the quotient of the infinite additive line $\mathbb{Z}$ by the periodicity subgroup $n\mathbb{Z}$.

The circle. The additive quotient $\mathbb{R}/\mathbb{Z}$ is naturally isomorphic to S^1 under $t + \mathbb{Z} \mapsto e^{2\pi it}$.

Parity quotient. $S_n/A_n \cong C_2$: after forgetting every even permutation, only parity remains.

2.5 The first isomorphism theorem

Theorem 2.8 (First isomorphism theorem). *If $\varphi : G \rightarrow H$ is a homomorphism, then*

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

Proof. Define

$$\bar{\varphi} : G/\ker \varphi \rightarrow \operatorname{im} \varphi, \quad g \ker \varphi \mapsto \varphi(g).$$

If $g \ker \varphi = h \ker \varphi$, then $h^{-1}g \in \ker \varphi$, so

$$e = \varphi(h^{-1}g) = \varphi(h)^{-1}\varphi(g),$$

and therefore $\varphi(g) = \varphi(h)$. Thus the map is well-defined. It is plainly a homomorphism and surjective onto the image. If $\bar{\varphi}(g \ker \varphi) = e$, then $g \in \ker \varphi$, so $g \ker \varphi$ is the identity coset. Hence the map is injective. $\square$

Deep intuition

This theorem says that a homomorphism forgets *exactly* its kernel. Once you deliberately collapse that lost information, no additional ambiguity remains, and the quotient becomes the image.

Proof architecture

When constructing a map out of a quotient, always ask first: *is it well-defined?* This is the central logical step. Surjectivity and injectivity are often routine afterward.

2.6 Group actions: turning abstract elements into motions

Definition 2.9. A left action of G on a set X is a map $G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x$, such that

$$e \cdot x = x, \quad (gh) \cdot x = g \cdot (h \cdot x).$$

For $x \in X$ define

$$\text{Orb}(x) = \{g \cdot x : g \in G\}, \quad \text{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

Deep intuition

A group action is a representation of abstract algebra as actual movement. The orbit answers “where can this object go?” The stabilizer answers “how much symmetry can happen without moving it?” These are complementary measurements: more possible destinations means fewer symmetries can keep the point fixed.

Theorem 2.10 (Orbit–stabilizer). *For any group action and $x \in X$, there is a bijection*

$$G/\text{Stab}(x) \longrightarrow \text{Orb}(x), \quad g\text{Stab}(x) \mapsto g \cdot x.$$

Hence, when G is finite,

$$|G| = |\text{Orb}(x)| |\text{Stab}(x)|.$$

Proof. If $g\text{Stab}(x) = h\text{Stab}(x)$, then $h^{-1}g \in \text{Stab}(x)$, so $(h^{-1}g) \cdot x = x$ and hence $g \cdot x = h \cdot x$; thus the map is well-defined. It is surjective by definition of the orbit. If $g \cdot x = h \cdot x$, then $(h^{-1}g) \cdot x = x$, so $h^{-1}g \in \text{Stab}(x)$ and the cosets are equal. Thus the map is injective. $\square$

Example 2.11 (Permuting two-element subsets). Let S_4 act on the six 2-element subsets of $\{1, 2, 3, 4\}$. The orbit of $\{1, 2\}$ is all six subsets. Its stabilizer consists of permutations that may swap 1, 2 and independently swap 3, 4, so has 4 elements. Orbit–stabilizer gives $24 = 6 \cdot 4$.

2.6.1 Conjugation and coordinate change

Every group acts on itself by

$$g \cdot x = gxg^{-1}.$$

The orbits are conjugacy classes.

Deep intuition

Conjugation is the algebraic form of changing coordinates. If x describes a symmetry in one labeling system and g changes the labeling, then gxg^{-1} is the same underlying symmetry written in the new coordinates. Conjugate elements are therefore not identical, but they encode the same intrinsic behavior.

Example 2.12 (Conjugacy in S_3). The conjugacy classes are

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

They correspond to cycle types $1 + 1 + 1$, $2 + 1$, and 3 . In S_n , two permutations are conjugate exactly when they have the same cycle type.

Why this matters later

This is one of the conceptual seeds of Chebotarev. A Frobenius element attached to a prime depends on a choice of prime upstairs; changing that choice conjugates the element. The conjugacy class, not the individual element, is the invariant that survives the choice.

2.7 Sylow theory: locating the largest p -power symmetry

Suppose $|G| = p^n m$ with $p \nmid m$. A subgroup of order p^n is called a Sylow p -subgroup.

Why this idea exists

Finite groups often look complicated because several prime-power phenomena are mixed together. Sylow theory separates one prime at a time. The p -part p^n is the largest amount of p -power symmetry that could possibly fit inside G .

Lemma 2.13. *If $p \nmid m$, then $\binom{p^n m}{p^n}$ is not divisible by p .*

Proof. Write

$$\binom{p^n m}{p^n} = m \prod_{j=1}^{p^n-1} \frac{p^n m - j}{p^n - j}.$$

For $1 \leq j < p^n$, write $j = p^r u$ with $p \nmid u$ and $r < n$. Then

$$p^n m - j = p^r (p^{n-r} m - u), \quad p^n - j = p^r (p^{n-r} - u).$$

Both parenthesized factors are nonzero modulo p , so numerator and denominator have the same p -adic valuation in each ratio. The total valuation is therefore $v_p(m) = 0$. $\square$

Theorem 2.14 (Sylow theorems). *Let $|G| = p^n m$ with $p \nmid m$. Then:*

1. G has a subgroup of order p^n ;
2. every p -subgroup is contained in a conjugate of a Sylow p -subgroup; in particular all Sylow p -subgroups are conjugate;
3. the number n_p of Sylow p -subgroups satisfies $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.

Proof. For existence, let X be the set of p^n -element subsets of G , with G acting by left translation. The lemma says $|X|$ is not divisible by p . Since X is a disjoint union of orbits, some orbit has size not divisible by p . Choose S in such an orbit and let $H = \text{Stab}(S)$. Then $[G : H] = |G \cdot S|$ is not divisible by p , so $p^n \mid |H|$. On the other hand, H acts freely on S by left multiplication: $hs = s$ implies $h = e$. Thus every H -orbit in S has size $|H|$, so $|H| \mid |S| = p^n$. Hence $|H| = p^n$.

Fix a Sylow subgroup P , and let H be any p -subgroup. Let H act on G/P by left multiplication. Every orbit size is a power of p , but $|G/P| = m$ is not divisible by p , so some orbit has size 1. Thus some coset gP is fixed by every $h \in H$, meaning $g^{-1}Hg \leq P$. Equivalently, $H \leq gPg^{-1}$. If H is itself Sylow, equality follows from the orders. Hence all Sylow subgroups are conjugate.

The conjugates of P are in bijection with $G/N_G(P)$, so $n_p = [G : N_G(P)]$. Because $P \leq N_G(P)$, this index divides $[G : P] = m$. Finally let P act by conjugation on the set of Sylow p -subgroups. A Sylow subgroup Q is fixed precisely when P normalizes Q . Then PQ is a subgroup and a p -group; it contains the Sylow subgroup P , so $PQ = P$, hence $Q \leq P$, and equality of orders gives $Q = P$. Thus P is the unique fixed point. Every other orbit has size divisible by p , so $n_p \equiv 1 \pmod{p}$. $\square$

Example ladder

Order $21 = 3 \cdot 7$. The number n_7 divides 3 and is 1 mod 7, so $n_7 = 1$. The Sylow 7-subgroup is normal. For n_3 , the constraints are $n_3 \mid 7$ and $n_3 \equiv 1 \pmod{3}$, so $n_3 \in \{1, 7\}$.

Order $15 = 3 \cdot 5$. $n_5 \mid 3$ and $n_5 \equiv 1 \pmod{5}$, so $n_5 = 1$; the 5-Sylow is normal. Also $n_3 \mid 5$ and $n_3 \equiv 1 \pmod{3}$, so $n_3 = 1$ or 4; divisibility eliminates 4, hence $n_3 = 1$. In fact every group of order 15 has both Sylow subgroups normal.

Rings, ideals, and quotient arithmetic

3.1 Rings as arithmetic environments

Definition 3.1. A commutative ring with identity is an abelian group $(R, +)$ equipped with associative, commutative multiplication, distributive over addition, and with a multiplicative identity 1.

Why this idea exists

A field is a world where every nonzero number can be divided by. A ring is what remains when division may fail. This is not a defect; it is exactly the setting in which divisibility, congruences, factorization, and prime ideals become visible.

Example ladder

$\mathbb{Z}$ is the basic ring: division usually leaves the ring. Polynomial rings $K[x]$ are rings in which the indeterminate x is not invertible. The quotient $\mathbb{Z}/n\mathbb{Z}$ is a ring whose zero-divisor behavior depends sharply on whether n is prime.

3.2 Ideals: the subsets that can be consistently declared zero

Definition 3.2. An additive subgroup $I \subseteq R$ is an *ideal* if

$$r \in R, a \in I \implies ra \in I.$$

Deep intuition

An ideal is not merely a smaller ring. It is a collection of quantities we are allowed to erase. If a is declared to be zero, then so must every multiple ra , because multiplication by ambient elements should respect equality. The absorption axiom is exactly the closure condition required for this erasure to be algebraically consistent.

Example ladder

In $\mathbb{Z}$. Every ideal is $n\mathbb{Z}$ for a unique $n \geq 0$.

Evaluation. In $K[x]$, the kernel of evaluation $f(x) \mapsto f(a)$ is $(x - a)$. Modding out by $(x - a)$ makes x literally equal to a .

Two variables. In $K[x, y]$, the ideal (x, y) consists of polynomials with zero constant term. Quotienting by it collapses both x and y to zero.

Proposition 3.3. *Every ideal of $\mathbb{Z}$ has the form $n\mathbb{Z}$ for a unique $n \geq 0$.*

Proof. Let I be an ideal. If $I = \{0\}$, take $n = 0$. Otherwise I contains a positive integer; let n be the least positive one. Certainly $n\mathbb{Z} \subseteq I$. For any $a \in I$, divide $a = qn + r$ with $0 \leq r < n$. Since $r = a - qn \in I$, minimality of n forces $r = 0$. Thus $a \in n\mathbb{Z}$, so $I = n\mathbb{Z}$. Uniqueness follows because if $n\mathbb{Z} = m\mathbb{Z}$ with $m, n \geq 0$, then each divides the other, hence $m = n$. $\square$

3.3 Quotient rings and the meaning of a relation

For an ideal I , define

$$R/I = \{a + I : a \in R\}$$

with

$$(a + I) + (b + I) = (a + b) + I, \quad (a + I)(b + I) = ab + I.$$

Proposition 3.4. *These operations are well-defined, and the quotient map $\pi : R \rightarrow R/I$, $a \mapsto a + I$, is a surjective ring homomorphism with kernel I .*

Proof. Suppose $a - a' \in I$ and $b - b' \in I$. Then

$$(a + b) - (a' + b') = (a - a') + (b - b') \in I.$$

For multiplication,

$$ab - a'b' = a(b - b') + b'(a - a') \in I,$$

using absorption by arbitrary elements of R . Hence both operations are independent of representatives. The remaining claims are immediate from the definitions. $\square$

Deep intuition

A quotient ring is an arithmetic world in which every element of I has been declared equal to zero. If $I = (f)$ in a polynomial ring, then the relation $f = 0$ becomes an algebraic law. This viewpoint is the bridge to adjoining algebraic numbers.

Example 3.5 (Making i by imposing $x^2 + 1 = 0$). In $\mathbb{R}[x]/(x^2 + 1)$, the class $\bar{x}$ satisfies $\bar{x}^2 = -1$. Every polynomial class has a unique representative $a + b\bar{x}$ because polynomial division by $x^2 + 1$ leaves a remainder of degree < 2 . The map

$$\mathbb{R}[x]/(x^2 + 1) \rightarrow \mathbb{C}, \quad a + b\bar{x} \mapsto a + bi$$

is an isomorphism. Thus $\mathbb{C}$ can be viewed as the real polynomial ring after imposing the relation $x^2 = -1$.

3.4 Kernels and the first isomorphism theorem for rings

Theorem 3.6. *If $\varphi : R \rightarrow S$ is a ring homomorphism, then*

$$R/\ker \varphi \cong \operatorname{im} \varphi.$$

Proof. Define $\bar{\varphi}(r + \ker \varphi) = \varphi(r)$. Exactly as for groups, equality of cosets implies the representatives differ by an element of the kernel, so the map is well-defined. It is a surjective ring homomorphism onto the image. Its kernel is the zero coset, so it is injective. $\square$

3.5 Prime ideals and maximal ideals

Definition 3.7. A proper ideal $\mathfrak{p} \subsetneq R$ is *prime* if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

A proper ideal $\mathfrak{m}$ is *maximal* if no ideal lies strictly between $\mathfrak{m}$ and R .

Deep intuition

A prime ideal makes “zero modulo $\mathfrak{p}$ ” behave like zero in an integral domain: a product can vanish only if one factor vanishes. A maximal ideal goes one step further: after quotienting, every nonzero class becomes invertible, so the quotient is a field.

Theorem 3.8 (Prime quotient criterion). *A proper ideal $\mathfrak{p}$ is prime if and only if $R/\mathfrak{p}$ is an integral domain.*

Proof. In the quotient,

$$(a + \mathfrak{p})(b + \mathfrak{p}) = 0 + \mathfrak{p} \iff ab \in \mathfrak{p}.$$

Thus the absence of zero divisors is exactly the condition that $ab \in \mathfrak{p}$ forces $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. $\square$

Theorem 3.9 (Maximal quotient criterion). *A proper ideal $\mathfrak{m}$ is maximal if and only if $R/\mathfrak{m}$ is a field.*

Proof. Assume $\mathfrak{m}$ maximal. If $a + \mathfrak{m} \neq 0$, then $a \notin \mathfrak{m}$. The ideal $\mathfrak{m} + (a)$ strictly contains $\mathfrak{m}$, hence equals R . So

$$1 = m + ra$$

for some $m \in \mathfrak{m}$, $r \in R$. Modulo $\mathfrak{m}$ this gives $(r + \mathfrak{m})(a + \mathfrak{m}) = 1 + \mathfrak{m}$, so every nonzero class is invertible.

Conversely, suppose $R/\mathfrak{m}$ is a field and $\mathfrak{m} \subsetneq I \subseteq R$. Then $I/\mathfrak{m}$ is an ideal of the field $R/\mathfrak{m}$, hence either 0 or the whole field. Therefore $I = \mathfrak{m}$ or $I = R$. $\square$

Example ladder

In $\mathbb{Z}$, (p) is maximal and prime exactly when p is prime. The ideal (0) is prime because $\mathbb{Z}$ is a domain, but not maximal because $\mathbb{Z}$ is not a field. In $K[x, y]$, (x) is prime because the quotient is $K[y]$, a domain; it is not maximal because $K[y]$ is not a field. The larger ideal (x, y) is maximal because the quotient is K .

3.6 The Chinese remainder theorem: independent coordinates

Definition 3.10. Ideals I, J are *comaximal* if $I + J = R$, equivalently if there exist $a \in I, b \in J$ with $a + b = 1$.

Why this idea exists

The Chinese remainder theorem says that congruence conditions can serve as independent coordinates when their moduli do not interfere. The algebraic meaning of “do not interfere” is comaximality.

Theorem 3.11 (Chinese remainder theorem). *If $I + J = R$, then*

$$R/(I \cap J) \cong R/I \times R/J.$$

Moreover $I \cap J = IJ$, so

$$R/IJ \cong R/I \times R/J.$$

Proof. Consider

$$\Phi : R \rightarrow R/I \times R/J, \quad r \mapsto (r + I, r + J).$$

Its kernel is $I \cap J$. Choose $a \in I, b \in J$ with $a + b = 1$. Given $(x + I, y + J)$, let

$$r = xb + ya.$$

Modulo I , we have $a \equiv 0$ and $b = 1 - a \equiv 1$, so $r \equiv x$. Modulo J , $b \equiv 0$ and $a \equiv 1$, so $r \equiv y$. Thus Φ is surjective, and the first isomorphism theorem gives the first claim.

Clearly $IJ \subseteq I \cap J$. If $z \in I \cap J$, then

$$z = z(a + b) = za + zb.$$

Because $z \in J$ and $a \in I$, $za \in IJ$; similarly $zb \in IJ$. Hence $z \in IJ$. □

Example 3.12 (The congruences 2, 3, 4 modulo 3, 5, 7). We seek x with

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 4 \pmod{7}.$$

Since 3, 5, 7 are pairwise coprime, the solution is unique modulo 105. One explicit solution is $x = 53$:

$$53 = 51 + 2 = 50 + 3 = 49 + 4.$$

The theorem explains uniqueness and existence simultaneously: the map

$$\mathbb{Z}/105\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$$

is an isomorphism.

Deep intuition

The CRT is best thought of as a coordinate change. A number modulo 105 can be encoded by three independent coordinates: its residues modulo 3, 5, and 7. Arithmetic can then be performed coordinatewise.

3.7 Correspondence of ideals

Theorem 3.13 (Ideal correspondence). *Let $\pi : R \rightarrow R/I$ be the quotient map. Ideals of R/I correspond bijectively to ideals J of R containing I , via*

$$J \longmapsto J/I, \quad \bar{J} \longmapsto \pi^{-1}(\bar{J}).$$

Proof. If $I \subseteq J$, then J/I is an ideal of R/I . Conversely, the inverse image of an ideal under a ring homomorphism is an ideal and necessarily contains $\ker \pi = I$. For $I \subseteq J$, one has $\pi^{-1}(J/I) = J$. For an ideal $\bar{J} \subseteq R/I$, surjectivity of π gives $\pi(\pi^{-1}(\bar{J})) = \bar{J}$. Thus the constructions are inverse. $\square$

Why this matters later

Quotients are the common structural idea running through groups, rings, and fields. In the next part, $K[x]/(m(x))$ will let us *manufacture* a field containing a root of m by declaring $m(x) = 0$.

Part III

Algebra II: fields, polynomials, and Galois symmetry

Field extensions and algebraic elements

4.1 Field extensions as enlargement of the allowed scalars

Definition 4.1. A *field extension* L/K means that K is a subfield of L . Then L is naturally a vector space over K . Its dimension, when finite, is the *degree*

$$[L : K] = \dim_K L.$$

Why this idea exists

Many equations have no solution in the field where they are first posed. The equation $x^2 = 2$ has no solution in $\mathbb{Q}$, but instead of giving up we can enlarge the number system just enough to contain a solution. Field extension theory asks: how much enlargement was necessary, and what new symmetries appear once the missing roots are present?

Deep intuition

Think of K as the vocabulary you are initially allowed to use. Passing to L adds new symbols, but the old arithmetic remains valid. The degree $[L : K]$ measures how many independent K -directions are needed to describe the larger field.

Example ladder

Quadratic extension. Every element of $\mathbb{Q}(\sqrt{2})$ has the form $a + b\sqrt{2}$, so $\{1, \sqrt{2}\}$ is a $\mathbb{Q}$ -basis and $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.

Complex numbers. $\mathbb{C} = \mathbb{R}(i)$ with basis $\{1, i\}$, hence $[\mathbb{C} : \mathbb{R}] = 2$.

A tower. $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$ lets us measure a complicated extension in stages.

Theorem 4.2 (Tower law). *If $K \subseteq E \subseteq L$ and both $[E : K]$ and $[L : E]$ are finite, then*

$$[L : K] = [L : E][E : K].$$

Proof. Let $e_1, \dots, e_r$ be a K -basis of E and $\ell_1, \dots, \ell_s$ an E -basis of L . We claim that the rs products $\ell_i e_j$ form a K -basis of L .

Every $x \in L$ can be written $x = \sum_i a_i \ell_i$ with $a_i \in E$, and each $a_i = \sum_j c_{ij} e_j$ with $c_{ij} \in K$, so the products span. For independence, if $\sum_{i,j} c_{ij} \ell_i e_j = 0$, regroup as

$$\sum_i \left(\sum_j c_{ij} e_j \right) \ell_i = 0.$$

Independence of the ℓ_i over E makes each parenthesis zero, and independence of the e_j over K then makes every $c_{ij} = 0$. Hence the products are a basis. $\square$

Deep intuition

Degrees multiply because “coordinates inside coordinates” multiply. If each element of L needs s coordinates from E , and each of those needs r coordinates from K , then L needs rs coordinates from K .

4.2 Algebraic elements and minimal polynomials

Definition 4.3. An element $\alpha \in L$ is *algebraic over K* if some nonzero $f \in K[x]$ satisfies $f(\alpha) = 0$. Otherwise α is *transcendental* over K .

Definition 4.4. If α is algebraic over K , its *minimal polynomial* $m_{\alpha,K}(x)$ is the unique monic polynomial of least positive degree in $K[x]$ with $m_{\alpha,K}(\alpha) = 0$.

Deep intuition

The minimal polynomial is the shortest algebraic “identity card” of α over the base field. Any polynomial relation satisfied by α must contain that identity card as a factor. Its degree is exactly the amount of linear information needed to adjoin α .

Proposition 4.5. *The minimal polynomial is irreducible. Moreover, if $f \in K[x]$ and $f(\alpha) = 0$, then $m_{\alpha,K} \mid f$.*

Proof. Divide $f = qm + r$ with $\deg r < \deg m$. Evaluating at α gives $r(\alpha) = 0$. By minimality, $r = 0$, so $m \mid f$. If $m = gh$ with both factors nonconstant, then $0 = m(\alpha) = g(\alpha)h(\alpha)$. Since L is a field, one factor vanishes, contradicting the minimal degree of m . $\square$

Theorem 4.6 (Simple algebraic extension as a quotient). *If α is algebraic over K with minimal polynomial m , then evaluation induces an isomorphism*

$$K[x]/(m) \cong K(\alpha), \quad f(x) + (m) \mapsto f(\alpha).$$

Consequently

$$[K(\alpha) : K] = \deg m,$$

and $1, \alpha, \dots, \alpha^{d-1}$ is a basis when $d = \deg m$.

Proof. The evaluation map $\text{ev}_\alpha : K[x] \rightarrow K(\alpha)$ has kernel consisting of all polynomials vanishing at α , which by the preceding proposition is precisely (m) . Its image is $K[\alpha]$. Because m is irreducible, $K[x]/(m)$ is a field; hence its image $K[\alpha]$ is a field containing K and α , so it equals $K(\alpha)$. The first isomorphism theorem gives the isomorphism. Polynomial division gives a unique representative of degree $< d$, yielding the stated basis. $\square$

Example 4.7 (Adjoining $\sqrt{2}$). For $\alpha = \sqrt{2}$, the polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$, so

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}).$$

Inside the quotient, the class $\bar{x}$ obeys $\bar{x}^2 = 2$. This is not merely notation: the quotient construction literally manufactures a field containing an element with square 2.

Example 4.8 (The minimal polynomial of $\sqrt{2} + \sqrt{3}$). Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\alpha^2 = 5 + 2\sqrt{6}, \quad \alpha^2 - 5 = 2\sqrt{6},$$

so

$$(\alpha^2 - 5)^2 = 24.$$

Thus α is a root of

$$f(x) = x^4 - 10x^2 + 1.$$

To prove this is minimal, note that

$$\sqrt{2} = \frac{\alpha^2 - 1}{2\alpha}, \quad \sqrt{3} = \frac{\alpha^2 + 1}{2\alpha},$$

so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Now $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, and $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$: if $\sqrt{3} = a + b\sqrt{2}$, squaring and comparing the irrational part forces $ab = 0$, leading to an impossible rational square equation. Thus

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4.$$

Hence the minimal polynomial has degree 4, so the monic degree-4 annihilating polynomial f is minimal.

Common pitfall

Producing an annihilating polynomial is not the same as finding the minimal polynomial. The annihilating polynomial may factor. You must still prove irreducibility or independently prove that no lower-degree relation can exist.

4.3 Irreducibility tools

Lemma 4.9 (Gauss's lemma). *The product of two primitive polynomials in $\mathbb{Z}[x]$ is primitive. Consequently, a primitive polynomial in $\mathbb{Z}[x]$ is reducible over $\mathbb{Q}$ if and only if it is reducible over $\mathbb{Z}$.*

Proof. A polynomial is primitive if the gcd of its coefficients is 1. Suppose primitive f, g had product whose coefficients were all divisible by a prime p . Reducing modulo p , we would have nonzero polynomials $\bar{f}, \bar{g} \in \mathbb{F}_p[x]$ with $\bar{f}\bar{g} = 0$, impossible because $\mathbb{F}_p[x]$ is an integral domain. Thus fg is primitive.

For the consequence, clear denominators in a factorization over $\mathbb{Q}$, then divide each integer factor by its content. Comparing contents and using the first part yields a factorization by primitive integer polynomials. The reverse implication is immediate. $\square$

Theorem 4.10 (Eisenstein's criterion). *Let*

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x].$$

Suppose a prime p satisfies $p \nmid a_n$, $p \mid a_i$ for $i < n$, and $p^2 \nmid a_0$. Then f is irreducible over $\mathbb{Q}$.

Proof. After dividing out the content, assume f primitive. If $f = gh$ nontrivially in $\mathbb{Z}[x]$, reduce modulo p . Since every non-leading coefficient vanishes mod p ,

$$\bar{f} = \bar{a}_n x^n.$$

Thus $\bar{g}$ and $\bar{h}$ must each be monomials, so their constant terms are divisible by p . Therefore the constant term $a_0 = g(0)h(0)$ is divisible by p^2 , contradiction. Gauss's lemma then excludes a factorization over $\mathbb{Q}$. $\square$

Example ladder

$x^3 - 2$ is Eisenstein at 2, hence irreducible over $\mathbb{Q}$. More generally, $x^n - p$ is Eisenstein at p for any prime p .

The criterion is sufficient, not necessary. For example $x^2 + x + 1$ is irreducible over $\mathbb{Q}$ but is not Eisenstein in its displayed form.

4.4 Splitting fields: containing the whole root picture

Definition 4.11. A *splitting field* of $f \in K[x]$ is a field extension L/K in which f factors completely into linear factors and which is generated over K by the roots of f .

Deep intuition

Adjoining one root answers a local question: “can I solve this equation at least once?” A splitting field answers the global question: “can I see the entire configuration of roots at once?” Galois symmetry only becomes fully visible when the whole root configuration is present.

Theorem 4.12 (Existence of splitting fields). *Every nonconstant $f \in K[x]$ has a splitting field.*

Proof. Induct on $\deg f$. Choose an irreducible factor p of f . The quotient $K[x]/(p)$ is a field containing a root α of p , hence of f . Over $K(\alpha)$, factor

$$f(x) = (x - \alpha)g(x).$$

By induction, g has a splitting field over $K(\alpha)$. That field contains α and all roots of g , hence all roots of f , and is generated by them. $\square$

Lemma 4.13 (Extending an embedding over one algebraic element). *Let $\sigma : K \hookrightarrow \Omega$ be a field embedding into an algebraically closed field Ω , and let α be algebraic over K with minimal polynomial $m(x) = \sum a_i x^i$. Define the transported polynomial*

$$\sigma(m)(x) = \sum \sigma(a_i) x^i.$$

For every root β of $\sigma(m)$ in Ω , there is a unique embedding

$$\tilde{\sigma} : K(\alpha) \hookrightarrow \Omega$$

extending σ and sending α to β .

Proof. Define on polynomials

$$\sum a_i \alpha^i \mapsto \sum \sigma(a_i) \beta^i.$$

If two polynomial expressions agree at α , their difference is divisible by m ; after applying σ to coefficients, the transported difference is divisible by $\sigma(m)$ and therefore vanishes at β . Thus the rule is well-defined. It is plainly a field homomorphism extending σ , and uniqueness follows because $K(\alpha)$ is generated by K and α . $\square$

Theorem 4.14 (Uniqueness of splitting fields). *Any two splitting fields of the same polynomial $f \in K[x]$ are isomorphic by an isomorphism fixing K .*

Proof. Let L and M be splitting fields inside algebraic closures. Starting with the identity on K , adjoin the roots generating L one at a time. At each step the extension lemma sends the next root to some root of the corresponding transported minimal polynomial; because M contains all roots of f , the embedding can be continued into M . We obtain a K -embedding $L \hookrightarrow M$. By symmetry there is a K -embedding $M \hookrightarrow L$. Both fields have finite degree over K (they are generated by finitely many algebraic roots), and the embeddings imply both degree inequalities, hence equal degrees. The embedding $L \hookrightarrow M$ is therefore onto and is an isomorphism. $\square$

4.5 Separable, normal, and Galois: distinct roots versus complete roots

Definition 4.15. A polynomial is *separable* if it has no repeated roots in a splitting field. An algebraic extension is separable if every element has a separable minimal polynomial.

Proposition 4.16 (Derivative criterion). *For $f \in K[x]$, the following are equivalent:*

1. f has a repeated root in a splitting field;
2. f and f' have a nonconstant common factor;
3. $\gcd(f, f') \neq 1$.

Proof. If α is a repeated root, write $f = (x - \alpha)^2 g$. Differentiating shows $(x - \alpha)$ divides f' , so f and f' share a factor. Conversely, if an irreducible polynomial p divides both f and f' , any root α of p is a common root of f and f' . In a factorization $f = (x - \alpha)^m g$ with $g(\alpha) \neq 0$, one has $f'(\alpha) = 0$ exactly when $m \geq 2$. The gcd formulation is equivalent to the existence of a nonconstant common factor. $\square$

Corollary 4.17. *Every irreducible polynomial over a field of characteristic 0 is separable.*

Proof. If irreducible f were inseparable, then f and f' would have a nonconstant gcd. Irreducibility forces $f \mid f'$, impossible unless $f' = 0$. In characteristic 0, a nonconstant polynomial cannot have zero derivative. Hence f is separable. $\square$

Definition 4.18. An algebraic extension L/K is *normal* if every irreducible $f \in K[x]$ having one root in L splits completely over L . A finite extension is *Galois* if it is both normal and separable.

Deep intuition

Separability and normality solve different problems. Separability says the conjugate roots are *distinct*: no root has collapsed onto itself with multiplicity. Normality says the extension is *complete*: if one conjugate is present, all its siblings are present. Galois means the root picture is both nondegenerate and complete, which is exactly the setting in which automorphisms can permute roots cleanly.

Example ladder

$\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is Galois: $x^2 - 2$ has two distinct roots $\pm\sqrt{2}$, both in the field.
 $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is separable (characteristic 0) but not normal: it contains the real root of $x^3 - 2$ but not the nonreal roots $\zeta_3 \sqrt[3]{2}$ and $\zeta_3^2 \sqrt[3]{2}$.
 In characteristic p , inseparability can genuinely occur. Over $K = \mathbb{F}_p(t)$, the polynomial $x^p - t$ has derivative 0. In an extension containing a p th root u of t , it becomes $(x - u)^p$, a single root with multiplicity p .

Proposition 4.19. *A finite extension L/K is normal if and only if it is the splitting field over K of some finite family of polynomials in $K[x]$.*

Proof. If L is the splitting field of a family, let $\sigma : L \hookrightarrow \overline{K}$ be a K -embedding. It sends each root of each polynomial to another root of the same polynomial, so $\sigma(L) \subseteq L$; equality follows from finite degree. Thus every K -embedding maps L to itself. If an irreducible polynomial has one root $\alpha \in L$, every conjugate root arises as the image of α under some extension of the embedding $K(\alpha) \hookrightarrow \overline{K}$; extending further to L and using $\sigma(L) = L$ shows every conjugate lies in L . Hence L/K is normal.

Conversely, if L/K is finite normal, choose generators $L = K(\alpha_1, \dots, \alpha_r)$ with each α_i algebraic. Let m_i be the minimal polynomial of α_i . Normality makes every m_i split in L , and L is generated by roots of the family $\{m_i\}$; hence it is their splitting field. $\square$

Galois groups, fixed fields, and finite-field Frobenius

5.1 Automorphisms as symmetries that preserve equations

Definition 5.1. For an extension L/K ,

$$\text{Aut}(L/K) = \{\sigma : L \rightarrow L : \sigma \text{ is a field automorphism and } \sigma|_K = \text{id}\}.$$

When L/K is Galois we write $\text{Gal}(L/K)$.

Proposition 5.2. *If $\sigma \in \text{Aut}(L/K)$, $f \in K[x]$, and $f(\alpha) = 0$, then $f(\sigma(\alpha)) = 0$. Hence $\sigma(\alpha)$ has the same minimal polynomial over K as α .*

Proof. Write $f(x) = \sum a_i x^i$ with $a_i \in K$. Then

$$0 = \sigma(f(\alpha)) = \sum \sigma(a_i) \sigma(\alpha)^i = \sum a_i \sigma(\alpha)^i = f(\sigma(\alpha)).$$

If m is the minimal polynomial of α , then $m(\sigma(\alpha)) = 0$, so the minimal polynomial of $\sigma(\alpha)$ divides m . Applying the same argument to σ^{-1} gives the reverse divisibility. $\square$

Deep intuition

A K -automorphism is forbidden to alter any equation written using coefficients from K . It may only rearrange the solutions. A Galois group is therefore a group of root permutations constrained by every algebraic relation among the roots.

Example 5.3 (Quadratic symmetry). In $L = \mathbb{Q}(\sqrt{2})$, every $\mathbb{Q}$ -automorphism must send $\sqrt{2}$ to another root of $x^2 - 2$, so either to $\sqrt{2}$ or $-\sqrt{2}$. Both choices define automorphisms. Thus

$$\text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) \cong C_2.$$

The nontrivial automorphism is the algebraic analogue of a reflection: it flips the sign of the new square root while fixing the rational axis.

5.2 Counting embeddings

Theorem 5.4 (Embedding count). *Let L/K be a finite separable extension and Ω an algebraic closure containing L . Then the number of K -embeddings $L \hookrightarrow \Omega$ is $[L : K]$.*

Proof. Choose a tower

$$K = K_0 \subset K_1 \subset \cdots \subset K_r = L, \quad K_j = K_{j-1}(\alpha_j).$$

Fix an embedding $\sigma : K_{j-1} \hookrightarrow \Omega$. The minimal polynomial m_j of α_j over K_{j-1} is separable, and the transported polynomial $\sigma(m_j)$ has exactly $[K_j : K_{j-1}]$ distinct roots in Ω . By the extension lemma, each root gives exactly one extension of σ to K_j . Thus at stage j the number of embeddings multiplies by $[K_j : K_{j-1}]$. Starting from the unique embedding of K fixing K and using the tower law gives

$$\prod_j [K_j : K_{j-1}] = [L : K].$$

□

Corollary 5.5. *If L/K is finite Galois, then*

$$|\mathrm{Gal}(L/K)| = [L : K].$$

Proof. Separability gives $[L : K]$ embeddings into an algebraic closure. Normality ensures every such embedding has image L , hence is an automorphism of L over K . □

5.3 Fixed fields and Artin's theorem

For a subgroup $H \leq \mathrm{Aut}(L)$, define

$$L^H = \{x \in L : \sigma(x) = x \text{ for every } \sigma \in H\}.$$

Deep intuition

The fixed field records what is invisible to a symmetry group. If H is a set of allowed motions, L^H is the collection of algebraic quantities that do not notice those motions at all. The bigger the symmetry group, the smaller the fixed field.

Lemma 5.6 (Dedekind independence). *Distinct field homomorphisms $\sigma_1, \dots, \sigma_n : L \rightarrow \Omega$ are linearly independent as functions over Ω . Thus if*

$$c_1\sigma_1(x) + \cdots + c_n\sigma_n(x) = 0 \quad \text{for all } x \in L,$$

then all $c_i = 0$.

Proof. Suppose a nontrivial relation exists with the smallest possible number r of nonzero coefficients. Relabel so $c_1, \dots, c_r \neq 0$. Since $\sigma_1 \neq \sigma_r$, choose $a \in L$ with $\sigma_1(a) \neq \sigma_r(a)$. For every x ,

$$0 = \sum_{i=1}^r c_i \sigma_i(ax) = \sum_{i=1}^r c_i \sigma_i(a) \sigma_i(x).$$

Subtract $\sigma_r(a)$ times the original relation. The σ_r term disappears, while the coefficient of σ_1 remains nonzero. This gives a nontrivial relation with fewer than r terms, contradiction. $\square$

Theorem 5.7 (Artin fixed-field theorem). *Let H be a finite group of automorphisms of a field L , and put $F = L^H$. Then*

$$[L : F] = |H|, \quad \text{Aut}(L/F) = H.$$

Proof. Write $H = \{\sigma_1, \dots, \sigma_n\}$. First we prove $[L : F] \leq n$. Take arbitrary $x_1, \dots, x_{n+1} \in L$. The homogeneous system

$$\sum_{j=1}^{n+1} \sigma_i(x_j) y_j = 0, \quad i = 1, \dots, n,$$

has a nonzero solution (y_j) over L because there are more unknowns than equations. Choose a solution with minimal support, relabel so $y_1 \neq 0$, and divide by y_1 so that $y_1 = 1$. For $\tau \in H$, applying τ to all equations yields another solution because left composition $\sigma_i \mapsto \tau \sigma_i$ permutes H . Therefore $(\tau(y_j) - y_j)$ is a solution whose first coordinate is zero. If it were nonzero, its support would be strictly smaller, contradicting minimality. Thus every y_j is fixed by every τ , so $y_j \in F$. The equation corresponding to the identity automorphism is then a nontrivial F -linear relation among the x_j . Hence any $n + 1$ elements are dependent and $[L : F] \leq n$.

Suppose $[L : F] = m < n$ and choose an F -basis $e_1, \dots, e_m$. The $n \times m$ matrix $(\sigma_i(e_j))$ has more rows than columns, so there are $c_i \in L$, not all zero, satisfying

$$\sum_{i=1}^n c_i \sigma_i(e_j) = 0 \quad \text{for every } j.$$

Every $x \in L$ is an F -linear combination of the e_j , and each σ_i fixes F , so $\sum_i c_i \sigma_i(x) = 0$ for every x . This contradicts Dedekind independence. Therefore $[L : F] = n$.

By definition $H \subseteq \text{Aut}(L/F)$. A finite extension of degree n has at most n distinct F -embeddings into an algebraic closure (the tower argument for the embedding count gives this upper bound even without separability). Hence $|\text{Aut}(L/F)| \leq n = |H|$, so equality holds. $\square$

5.4 The fundamental theorem of Galois theory

Theorem 5.8 (Finite Galois correspondence). *Let L/K be finite Galois and $G = \text{Gal}(L/K)$. Then*

$$\{K \subseteq E \subseteq L\} \longleftrightarrow \{H \leq G\}$$

is an inclusion-reversing bijection

$$E \mapsto \text{Gal}(L/E), \quad H \mapsto L^H.$$

Moreover,

$$[L : E] = |\text{Gal}(L/E)|, \quad [E : K] = [G : \text{Gal}(L/E)].$$

Finally, E/K is Galois if and only if $\text{Gal}(L/E) \triangleleft G$; then restriction induces

$$G / \text{Gal}(L/E) \cong \text{Gal}(E/K).$$

Proof. Let E be intermediate and $H = \text{Gal}(L/E)$. Since L/K is Galois, L/E is finite Galois: separability remains true over the larger base field, and a splitting-field presentation of L/K is still a splitting-field presentation after adjoining E . Thus $|H| = [L : E]$. Artin's theorem gives $[L : L^H] = |H|$, while $E \subseteq L^H$. Equality of degrees forces $E = L^H$.

Conversely, for $H \leq G$, Artin's theorem says $\text{Aut}(L/L^H) = H$, hence $\text{Gal}(L/L^H) = H$. Thus the assignments are inverse. Inclusion reversal is immediate: more elements required to be fixed means fewer automorphisms survive.

The degree formulas follow from $[L : E] = |H|$ and $[L : K] = |G|$ together with the tower law.

Now let $H = \text{Gal}(L/E)$. For $\sigma \in G$ one checks

$$\sigma H \sigma^{-1} = \text{Gal}(L/\sigma(E)).$$

Thus H is normal exactly when $\sigma(E) = E$ for every $\sigma \in G$. If E/K is Galois, every K -embedding of E maps E to itself, so this invariance holds. Conversely, if H is normal, every $\sigma \in G$ maps $E = L^H$ to itself; every K -embedding of E extends to a K -embedding of L because L/K is normal and separable, so all embeddings of E land in E . Hence E/K is normal; it is separable as a subextension of the separable extension L/K , so it is Galois.

When H is normal, restriction $G \rightarrow \text{Gal}(E/K)$ is a homomorphism with kernel H . It is surjective because every K -embedding of E into an algebraic closure extends to an automorphism of L and, since E/K is normal, maps E to itself. The first isomorphism theorem gives the quotient isomorphism. $\square$

Deep intuition

The Galois correspondence reverses size because “more field” means “less symmetry.” Enlarging E adds algebraic quantities that automorphisms are required to fix, so the subgroup gets smaller. Conversely, a larger subgroup moves more things, so fewer elements remain fixed.

5.5 A complete concrete lattice: $\mathbb{Q}(\sqrt{2}, \sqrt{3})$

Let

$$L = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

The degree is 4. Every automorphism may independently choose the signs of $\sqrt{2}$ and $\sqrt{3}$, because those choices preserve their defining equations and extend to the whole field. Thus

$$G \cong C_2 \times C_2.$$

Write

$$\begin{aligned}\sigma(\sqrt{2}) &= -\sqrt{2}, & \sigma(\sqrt{3}) &= \sqrt{3}, \\ \tau(\sqrt{2}) &= \sqrt{2}, & \tau(\sqrt{3}) &= -\sqrt{3}.\end{aligned}$$

Then the subgroup–field dictionary is

subgroup	fixed field
G	$\mathbb{Q}$
$\langle \sigma \rangle$	$\mathbb{Q}(\sqrt{3})$
$\langle \tau \rangle$	$\mathbb{Q}(\sqrt{2})$
$\langle \sigma\tau \rangle$	$\mathbb{Q}(\sqrt{6})$
$\{1\}$	L

The “mysterious” third quadratic field $\mathbb{Q}(\sqrt{6})$ appears because the simultaneous sign flip fixes the product $\sqrt{2}\sqrt{3} = \sqrt{6}$.

Deep intuition

A fixed field can often be guessed by looking for expressions whose sign changes cancel. This is a useful practical way to discover intermediate fields before verifying them by degrees.

5.6 Worked splitting field: $x^3 - 2$

Let $\alpha = \sqrt[3]{2}$ and $\zeta = e^{2\pi i/3}$. The three roots are

$$\alpha, \quad \zeta\alpha, \quad \zeta^2\alpha,$$

so the splitting field is $L = \mathbb{Q}(\alpha, \zeta)$. Eisenstein at 2 gives $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(\alpha) \subset \mathbb{R}$ while $\zeta \notin \mathbb{R}$, adjoining ζ has degree 2. Thus $[L : \mathbb{Q}] = 6$.

The extension is the splitting field of a separable polynomial, hence Galois. Its group has order 6 and acts faithfully on the three roots. Therefore

$$\text{Gal}(L/\mathbb{Q}) \hookrightarrow S_3.$$

Both groups have order 6, so

$$\text{Gal}(L/\mathbb{Q}) \cong S_3.$$

Complex conjugation fixes α and swaps the other two roots, so it corresponds to a transposition. Its fixed field has degree 3, and $\mathbb{Q}(\alpha)$ is visibly fixed, so it is exactly $\mathbb{Q}(\alpha)$.

Why this matters later

This example will return in the synthesis chapter. Modulo different primes, $x^3 - 2$ factors in patterns $1 + 1 + 1$, $1 + 2$, or 3 . Those factor degrees match the three cycle types in S_3 , foreshadowing the Frobenius classes counted by Chebotarev.

5.7 Finite fields from first principles

Why this idea exists

Finite fields are the local arithmetic worlds produced by reducing algebraic integers modulo prime ideals. Their canonical automorphism, Frobenius, is the local symmetry that Chebotarev later lifts back into a global Galois group.

Lemma 5.9 (Freshman's dream). *If K has characteristic $p > 0$, then*

$$(a + b)^p = a^p + b^p.$$

Hence $x \mapsto x^p$ is a ring homomorphism.

Proof. The binomial theorem gives

$$(a + b)^p = a^p + b^p + \sum_{k=1}^{p-1} \binom{p}{k} a^k b^{p-k}.$$

For $0 < k < p$, the prime p divides $\binom{p}{k}$, so every middle coefficient vanishes in characteristic p . Multiplicativity is $(ab)^p = a^p b^p$. $\square$

Theorem 5.10 (Existence and uniqueness of finite fields). *For every prime power $q = p^r$ there exists a field with exactly q elements, and any two such fields are isomorphic. It is denoted $\mathbb{F}_q$.*

Proof. Let L be a splitting field over $\mathbb{F}_p$ of $x^q - x$, and let

$$S = \{a \in L : a^q = a\}$$

be its set of roots. Because q is a power of p , repeated use of the freshman's dream gives

$$(a + b)^q = a^q + b^q,$$

so S is closed under addition and subtraction; it is closed under multiplication because $(ab)^q = a^q b^q$. If $a \neq 0$ lies in S , then $(a^{-1})^q = (a^q)^{-1} = a^{-1}$, so S is a field. The derivative of $x^q - x$ is $qx^{q-1} - 1 = -1$ in characteristic p , so the polynomial has no repeated roots. Since it splits in L and has degree q , it has exactly q distinct roots. Thus $|S| = q$.

For uniqueness, let E be any field with q elements. Every nonzero $a \in E$ satisfies $a^{q-1} = 1$ by Lagrange's theorem applied to $E^\times$, so every a satisfies $a^q = a$. Therefore E is a splitting field over $\mathbb{F}_p$ of $x^q - x$. Uniqueness of splitting fields makes any two such fields isomorphic. $\square$

Example 5.11 (The field $\mathbb{F}_4$). The polynomial $x^2 + x + 1$ has no root in $\mathbb{F}_2$, hence is irreducible. Therefore

$$\mathbb{F}_4 \cong \mathbb{F}_2[u]/(u^2 + u + 1).$$

The elements are $0, 1, u, u + 1$, with $u^2 = u + 1$ because $u^2 + u + 1 = 0$ in characteristic 2. The Frobenius map is squaring:

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad u \mapsto u^2 = u + 1, \quad u + 1 \mapsto (u + 1)^2 = u^2 + 1 = u.$$

So Frobenius fixes the base field $\mathbb{F}_2$ and swaps the two new conjugate elements. This four-element example contains the entire finite-field philosophy in miniature.

5.8 Frobenius automorphism

Theorem 5.12 (Finite-field Frobenius). *Let $q = p^r$. On $\mathbb{F}_{q^n}$ the map*

$$F(x) = x^q$$

is an $\mathbb{F}_q$ -automorphism of exact order n , and

$$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle F \rangle \cong C_n.$$

Proof. The map $x \mapsto x^q$ is obtained by iterating $x \mapsto x^p$, so it is a homomorphism. It is injective because $x^q = y^q$ implies $(x - y)^q = 0$, hence $x = y$; on a finite set it is therefore surjective. Every $a \in \mathbb{F}_q$ satisfies $a^q = a$, so F fixes the base field.

For $x \in \mathbb{F}_{q^n}$,

$$F^n(x) = x^{q^n} = x,$$

so the order of F divides n . Since $|\mathbb{F}_{q^n}| = q^n$ and $|\mathbb{F}_q| = q$, viewing $\mathbb{F}_{q^n}$ as a vector space over $\mathbb{F}_q$ gives $[\mathbb{F}_{q^n} : \mathbb{F}_q] = n$. If $F^d = \text{id}$ for some $0 < d < n$, then all q^n elements of $\mathbb{F}_{q^n}$ would be roots of $x^{q^d} - x$, a nonzero polynomial of degree $q^d < q^n$, contradicting the root bound. Thus F has exact order n .

The polynomial $x^{q^n} - x$ has all elements of $\mathbb{F}_{q^n}$ as roots and derivative -1 , so $\mathbb{F}_{q^n}$ is its splitting field over $\mathbb{F}_q$ and the extension is Galois. Its Galois group has order n . The powers $1, F, \dots, F^{n-1}$ already give n distinct automorphisms, so they form the whole group. $\square$

Deep intuition

Frobenius is a clock. Repeatedly raise a root to the q th power and you walk through its conjugates; the first time you return records the degree of the irreducible polynomial that created the root.

Theorem 5.13 (Frobenius orbit and irreducible degree). *Let $f \in \mathbb{F}_q[x]$ be irreducible of degree d , and let α be a root. Then its roots are exactly*

$$\alpha, \alpha^q, \alpha^{q^2}, \dots, \alpha^{q^{d-1}},$$

and d is the smallest positive integer with $\alpha^{q^d} = \alpha$.

Proof. The field $\mathbb{F}_q(\alpha)$ has degree d , hence q^d elements and is isomorphic to $\mathbb{F}_{q^d}$. Therefore $\alpha^{q^d} = \alpha$. Since Frobenius fixes the coefficients of f , every α^{q^j} is also a root of f .

Let e be the smallest positive return time. Then the e points

$$\alpha, \alpha^q, \dots, \alpha^{q^{e-1}}$$

are distinct. Their orbit polynomial

$$g(x) = \prod_{j=0}^{e-1} (x - \alpha^{q^j})$$

is fixed by Frobenius coefficientwise because Frobenius cyclically permutes its roots. The fixed elements of Frobenius in an algebraic closure satisfy $c^q = c$, so they are precisely $\mathbb{F}_q$; hence $g \in \mathbb{F}_q[x]$. Since $g(\alpha) = 0$, the minimal polynomial f divides g , giving $d \leq e$. But $\alpha^{q^d} = \alpha$, so $e \leq d$. Therefore $e = d$, and the listed points are all roots of f . $\square$

Why this matters later

Over a finite field, an irreducible factor of degree d corresponds to a Frobenius cycle of length d on its roots. Later, reducing a polynomial modulo a prime turns factorization type into cycle type of a Frobenius element in a Galois group.

Part IV

Real analysis: controlling approximation

Sequences, completeness, and infinite series

6.1 Limits as eventual accuracy

Definition 6.1. A sequence (a_n) of real or complex numbers converges to L if

$$\forall \varepsilon > 0 \exists N \forall n \geq N : |a_n - L| < \varepsilon.$$

We write $a_n \rightarrow L$.

Why this idea exists

A limit is not the claim that the terms eventually equal L . It is a promise about arbitrary accuracy: no matter how fine a tolerance you request, the tail of the sequence eventually stays inside it forever.

Deep intuition

Imagine a microscope centered at L . You are allowed to increase its magnification by choosing a smaller ε . The sequence converges exactly when, after each increase in magnification, there is some stage beyond which every later term remains in view. The index N is the price paid for the requested resolution.

Example 6.2 (The sequence $1/n$). Given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then for $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Thus $1/n \rightarrow 0$.

Example 6.3 (Why $(-1)^n$ does not converge). Suppose it converged to L . Choose $\varepsilon = 1/2$. Eventually all terms would lie within $1/2$ of L . But arbitrarily far out there are both 1 and -1 , which are distance 2 apart and cannot both lie in the same interval of radius $1/2$. Contradiction.

Proposition 6.4 (Uniqueness of limits). *A convergent sequence has a unique limit.*

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow M$. If $L \neq M$, choose $\varepsilon = |L - M|/3$. For all sufficiently large n ,

$$|a_n - L| < \varepsilon, \quad |a_n - M| < \varepsilon.$$

Then the triangle inequality gives

$$|L - M| \leq |L - a_n| + |a_n - M| < 2|L - M|/3,$$

impossible. □

Proposition 6.5 (Algebra of limits). *If $a_n \rightarrow a$ and $b_n \rightarrow b$, then $a_n + b_n \rightarrow a + b$ and $a_n b_n \rightarrow ab$. If $b \neq 0$ and $b_n \neq 0$ eventually, then $a_n/b_n \rightarrow a/b$.*

Proof. For sums,

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b|,$$

so make each term $< \varepsilon/2$. For products, convergent sequences are bounded: choose N so $|a_n - a| < 1$ eventually, hence $|a_n| \leq |a| + 1$. Then

$$|a_n b_n - ab| \leq |a_n| |b_n - b| + |b| |a_n - a|,$$

and choose the two errors small enough. For quotients, $b_n \rightarrow b \neq 0$ implies $|b_n| \geq |b|/2$ eventually, so

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b_n - b|}{|b_n b|} \leq \frac{2}{|b|^2} |b_n - b| \rightarrow 0,$$

and multiply by a_n . □

6.2 Cauchy sequences: detecting convergence from inside

Definition 6.6. A sequence (a_n) is *Cauchy* if

$$\forall \varepsilon > 0 \exists N \forall m, n \geq N : |a_m - a_n| < \varepsilon.$$

Why this idea exists

The definition of convergence mentions the limit L , which may be unknown. A Cauchy sequence replaces comparison with an unknown destination by comparison among sufficiently late terms. It is an internal convergence detector.

Deep intuition

A convergent sequence is a group of travelers approaching the same destination. Far enough into the journey, they must all be close to one another even if we do not know where the destination is. Completeness of the ambient number system says the converse: if the travelers bunch together indefinitely, there really is a destination inside the space.

Proposition 6.7. *Every convergent sequence is Cauchy.*

Proof. If $a_n \rightarrow L$, choose N so $|a_n - L| < \varepsilon/2$ for $n \geq N$. Then for $m, n \geq N$,

$$|a_m - a_n| \leq |a_m - L| + |a_n - L| < \varepsilon.$$

□

Lemma 6.8. *Every Cauchy sequence is bounded.*

Proof. Choose N so $|a_n - a_N| < 1$ for all $n \geq N$. Then the tail lies in the bounded interval $[a_N - 1, a_N + 1]$. The finitely many earlier terms are also bounded, so the whole sequence is bounded. □

6.3 Completeness of the real numbers

We take as foundational the least-upper-bound property: every nonempty subset of $\mathbb{R}$ bounded above has a supremum.

Lemma 6.9 (Bolzano–Weierstrass). *Every bounded real sequence has a convergent subsequence.*

Proof. Suppose $a_n \in [A, B]$. Bisect $[A, B]$. At least one half contains infinitely many sequence terms; choose such a half and call it I_1 . Bisect I_1 and choose a half I_2 containing infinitely many terms. Continue, obtaining nested closed intervals I_k of length $(B - A)/2^k$, each containing infinitely many terms.

Choose indices $n_1 < n_2 < \dots$ with $a_{n_k} \in I_k$. Let ℓ_k be the left endpoint of I_k . The increasing bounded sequence ℓ_k has supremum L . Because the interval lengths tend to 0 and $L \in I_k$ for every k , both a_{n_k} and L lie in I_k , so

$$|a_{n_k} - L| \leq |I_k| \rightarrow 0.$$

Thus $a_{n_k} \rightarrow L$. □

Theorem 6.10 (Completeness of $\mathbb{R}$). *Every Cauchy sequence of real numbers converges.*

Proof. A Cauchy sequence is bounded, so Bolzano–Weierstrass gives a convergent subsequence $a_{n_k} \rightarrow L$. Given $\varepsilon > 0$, choose N so $|a_m - a_n| < \varepsilon/2$ whenever $m, n \geq N$. Choose k so $n_k \geq N$ and $|a_{n_k} - L| < \varepsilon/2$. Then for every $n \geq N$,

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \varepsilon.$$

Hence the full sequence converges to L . □

Example ladder

The rational numbers are not complete. Decimal truncations of $\sqrt{2}$, viewed as rational numbers, form a Cauchy sequence in $\mathbb{Q}$ but have no rational limit. Completeness is the statement that $\mathbb{R}$ has no such “missing points.”

6.4 Infinite series as convergence of accumulated totals

Definition 6.11. The series $\sum_{n=1}^{\infty} a_n$ converges if its partial sums

$$s_N = \sum_{n=1}^N a_n$$

converge. It converges *absolutely* if $\sum |a_n|$ converges.

Deep intuition

A series is not an infinite act of addition performed all at once. It is a sequence of finite totals. Every series question is therefore a disguised sequence question about the partial sums.

Theorem 6.12 (Cauchy criterion for series). *The series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is N such that*

$$\left| \sum_{k=m}^n a_k \right| < \varepsilon \quad \text{whenever } n \geq m \geq N.$$

Proof. The difference of partial sums is

$$s_n - s_{m-1} = \sum_{k=m}^n a_k.$$

Thus the displayed condition is exactly the Cauchy condition for the partial-sum sequence. Completeness of $\mathbb{R}$ or $\mathbb{C}$ converts Cauchy into convergence. $\square$

Corollary 6.13. *Absolute convergence implies convergence.*

Proof. If $\sum |a_n|$ converges, then for large $m \leq n$ its tail satisfies

$$\sum_{k=m}^n |a_k| < \varepsilon.$$

By the triangle inequality,

$$\left| \sum_{k=m}^n a_k \right| \leq \sum_{k=m}^n |a_k| < \varepsilon.$$

The Cauchy criterion gives convergence. $\square$

A necessary condition for $\sum a_n$ to converge is $a_n \rightarrow 0$, because $a_n = s_n - s_{n-1}$. It is not sufficient.

6.5 The harmonic series and the scale of divergence

Theorem 6.14. *The harmonic series $\sum_{n=1}^{\infty} 1/n$ diverges.*

Proof. Group terms in dyadic blocks:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots.$$

The block from $2^{k-1} + 1$ through 2^k has 2^{k-1} terms, each at least $1/2^k$, so each block contributes at least $1/2$. Hence partial sums exceed $1 + k/2$ after k blocks and are unbounded. $\square$

Deep intuition

The terms $1/n$ do go to zero, but too slowly. Convergence depends not only on whether individual contributions vanish, but on whether the total mass remaining in the tail can be made small.

6.6 Comparison, integral, ratio, and root tests

Theorem 6.15 (Comparison test). *Suppose $0 \leq a_n \leq b_n$ eventually. If $\sum b_n$ converges, then $\sum a_n$ converges. If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

Proof. For nonnegative terms, partial sums are increasing. If $\sum b_n$ converges, its partial sums are bounded, and the partial sums of $\sum a_n$ are eventually bounded by them; hence they converge by monotone completeness. The second statement is the contrapositive. $\square$

Theorem 6.16 (p -series). *The series $\sum_{n=1}^{\infty} n^{-p}$ converges exactly when $p > 1$.*

Proof. For $p \leq 0$, the terms do not tend to zero. For $0 < p \leq 1$, $n^{-p} \geq n^{-1}$, so the harmonic comparison gives divergence. If $p > 1$, group into dyadic blocks. For $2^k \leq n < 2^{k+1}$, $n^{-p} \leq 2^{-kp}$ and there are 2^k terms, so the block sum is at most $2^{k(1-p)}$. The geometric series $\sum 2^{k(1-p)}$ converges because $p > 1$. $\square$

Theorem 6.17 (Integral test). *If $f : [1, \infty) \rightarrow [0, \infty)$ is continuous and decreasing, then*

$$\sum_{n=1}^{\infty} f(n)$$

converges if and only if $\int_1^{\infty} f(x) dx$ converges.

Proof. For decreasing f and $n \geq 1$,

$$f(n+1) \leq \int_n^{n+1} f(x) dx \leq f(n).$$

Summing from $n = 1$ to N gives

$$\sum_{n=2}^{N+1} f(n) \leq \int_1^{N+1} f(x) dx \leq \sum_{n=1}^N f(n).$$

Thus the integral and series partial sums bound one another up to a fixed first term, so they converge or diverge together. $\square$

Example 6.18. For $n \geq 2$,

$$\sum \frac{1}{n(\log n)^2}$$

converges because

$$\int_2^{\infty} \frac{dx}{x(\log x)^2} = \left[-\frac{1}{\log x} \right]_2^{\infty} = \frac{1}{\log 2}.$$

The nearby series $\sum 1/(n \log n)$ diverges because its integral is $\log \log x$.

Theorem 6.19 (Ratio test). *If $a_n \neq 0$ eventually and*

$$L = \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1,$$

then $\sum a_n$ converges absolutely. If

$$\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1,$$

then $a_n \not\rightarrow 0$ and the series diverges.

Proof. If $L < 1$, choose r with $L < r < 1$. Eventually $|a_{n+1}| \leq r|a_n|$, so by induction $|a_{N+k}| \leq |a_N|r^k$. The tail is dominated by a geometric series. If the $\liminf$ is > 1 , choose $r > 1$ below it. Eventually $|a_{n+1}| \geq r|a_n|$, so the magnitudes cannot tend to zero. $\square$

Theorem 6.20 (Root test). *Let*

$$L = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

If $L < 1$, the series $\sum a_n$ converges absolutely. If $L > 1$, it diverges.

Proof. If $L < 1$, choose r with $L < r < 1$. Eventually $|a_n|^{1/n} \leq r$, so $|a_n| \leq r^n$ and geometric comparison applies. If $L > 1$, choose r with $1 < r < L$. Infinitely many n satisfy $|a_n|^{1/n} > r$, hence $|a_n| > r^n$ infinitely often; in particular $a_n \not\rightarrow 0$. $\square$

Deep intuition

The ratio and root tests ask whether terms eventually behave geometrically. Geometric decay is the canonical convergent scale: once you can dominate the tail by Cr^n with $r < 1$, the problem is essentially solved.

Theorem 6.21 (Alternating series test). *If $b_n \geq 0$ decreases to 0, then*

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n$$

converges, and the error after N terms has magnitude at most b_{N+1} .

Proof. The even partial sums satisfy

$$s_{2k+2} - s_{2k} = b_{2k+1} - b_{2k+2} \geq 0,$$

so they increase. The odd partial sums decrease, and $s_{2k} \leq s_{2k+1}$. The gap $s_{2k+1} - s_{2k} = b_{2k+1} \rightarrow 0$, so the two monotone subsequences converge to the same limit. The full sequence of partial sums converges. The limit lies between successive partial sums, giving the error bound. $\square$

Example ladder

The alternating harmonic series $\sum (-1)^{n+1}/n$ converges but not absolutely. This is a concrete reminder that cancellation can create convergence even when the total magnitude is infinite.

Uniform convergence and power series

7.1 Pointwise versus uniform convergence

Definition 7.1. A sequence of functions $f_n : X \rightarrow \mathbb{R}$ or $\mathbb{C}$ converges *pointwise* to f if

$$\forall x \in X \ \forall \varepsilon > 0 \ \exists N = N(x, \varepsilon) \ \forall n \geq N : |f_n(x) - f(x)| < \varepsilon.$$

It converges *uniformly* if

$$\forall \varepsilon > 0 \ \exists N = N(\varepsilon) \ \forall n \geq N \ \forall x \in X : |f_n(x) - f(x)| < \varepsilon.$$

Equivalently,

$$\sup_{x \in X} |f_n(x) - f(x)| \rightarrow 0.$$

Deep intuition

Pointwise convergence lets every point negotiate its own schedule. Uniform convergence imposes one global deadline: after a single N , every point must be accurate simultaneously. The difference is entirely a difference in quantifier order, but it has major analytic consequences.

Example 7.2 (The standard boundary-layer example). On $[0, 1]$, let $f_n(x) = x^n$. For $0 \leq x < 1$, $x^n \rightarrow 0$, while $f_n(1) = 1$. Thus the pointwise limit is

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

The convergence is not uniform. Indeed, for $x_n = (1/2)^{1/n} < 1$,

$$|f_n(x_n) - f(x_n)| = x_n^n = \frac{1}{2}.$$

So the uniform error is at least $1/2$ for every n .

On every smaller interval $[0, a]$ with $a < 1$, however,

$$\sup_{0 \leq x \leq a} |x^n| = a^n \rightarrow 0,$$

so convergence is uniform there.

Deep intuition

The problem with x^n is a moving boundary layer near $x = 1$. For each fixed $x < 1$, the spike eventually moves past you; but no single time works for points that keep moving closer to 1. Pointwise convergence can hide errors that migrate through the domain.

Example 7.3 (A moving spike). Define on $[0, 1]$

$$f_n(x) = \max\{1 - n|x - 1/n|, 0\}.$$

For each fixed $x > 0$, the spike eventually lies near 0 and misses x , while $f_n(0) = 0$ for all n , so $f_n \rightarrow 0$ pointwise. But $f_n(1/n) = 1$, hence

$$\sup_x |f_n(x)| = 1$$

for all n . This makes the “moving error” phenomenon visible geometrically.

7.2 Uniform limits preserve continuity

Theorem 7.4. *If $f_n : X \rightarrow \mathbb{R}$ or $\mathbb{C}$ are continuous on a metric space and $f_n \rightarrow f$ uniformly, then f is continuous.*

Proof. Fix x_0 and $\varepsilon > 0$. Uniform convergence gives N such that

$$|f_N(x) - f(x)| < \varepsilon/3$$

for every x . Continuity of f_N at x_0 gives $\delta > 0$ such that $d(x, x_0) < \delta$ implies

$$|f_N(x) - f_N(x_0)| < \varepsilon/3.$$

Then

$$|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < \varepsilon.$$

□

Proof architecture

The proof uses an “approximate by one good continuous function” strategy. Uniform convergence is what lets the same f_N approximate f at both x and x_0 simultaneously.

7.3 Uniform convergence and integration

Theorem 7.5. *If continuous $f_n : [a, b] \rightarrow \mathbb{R}$ converge uniformly to f , then*

$$\int_a^b f_n(x) dx \longrightarrow \int_a^b f(x) dx.$$

Proof. The uniform limit is continuous, hence integrable. Moreover,

$$\left| \int_a^b (f_n - f) \right| \leq \int_a^b |f_n - f| \leq (b - a) \|f_n - f\|_\infty \rightarrow 0.$$

□

Deep intuition

Integration averages error over a finite interval. If the pointwise error is everywhere at most δ , the total accumulated error is at most interval length times δ .

7.4 Function series and the Weierstrass M-test

Theorem 7.6 (Weierstrass M-test). *Suppose $|f_n(x)| \leq M_n$ for every $x \in X$, and $\sum M_n$ converges. Then $\sum f_n$ converges uniformly and absolutely on X .*

Proof. For $n \geq m$,

$$\sup_{x \in X} \left| \sum_{k=m}^n f_k(x) \right| \leq \sum_{k=m}^n M_k.$$

The right-hand side tends uniformly to 0 as $m, n \rightarrow \infty$ because $\sum M_k$ is Cauchy. Thus the partial sums are uniformly Cauchy. Since $\mathbb{R}$ and $\mathbb{C}$ are complete, they converge pointwise to a function, and the same estimate gives uniform convergence. Absolute convergence follows from $\sum |f_n(x)| \leq \sum M_n$. □

Deep intuition

The M-test replaces a complicated function problem by a scalar bookkeeping problem. Find one numerical envelope M_n that controls the n th function everywhere; then convergence of the envelope pays for uniform convergence of the entire series.

7.5 Why differentiation is more delicate

Uniform convergence alone does not preserve derivatives.

Example 7.7. Let

$$f_n(x) = \frac{\sin(nx)}{n}$$

on $\mathbb{R}$. Then $\|f_n\|_\infty \leq 1/n$, so $f_n \rightarrow 0$ uniformly. But

$$f'_n(x) = \cos(nx)$$

does not converge pointwise for many x , let alone uniformly. Differentiation magnifies high-frequency oscillation.

Theorem 7.8 (Uniform convergence of derivatives). *Let $f_n : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. Suppose:*

1. $f'_n(x)$ converges uniformly on $[a, b]$ to a function $g(x)$;
2. for some $x_0 \in [a, b]$, the numbers $f_n(x_0)$ converge.

Then f_n converges uniformly to a differentiable function f , and $f' = g$.

Proof. By the fundamental theorem of calculus,

$$f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n(t) dt.$$

The first terms converge by hypothesis, and uniform convergence of f'_n makes the integrals converge uniformly to $\int_{x_0}^x g(t) dt$. Therefore f_n converges uniformly to

$$f(x) = c + \int_{x_0}^x g(t) dt,$$

where $c = \lim f_n(x_0)$. The uniform limit g is continuous, so the fundamental theorem of calculus gives $f' = g$. $\square$

Deep intuition

Knowing derivatives controls changes, but it does not fix an additive constant. That is why one anchor value $f_n(x_0)$ is required. “Derivative plus one basepoint” reconstructs the function.

7.6 Power series and the radius of convergence

Consider

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

Theorem 7.9 (Cauchy–Hadamard). *Let*

$$L = \limsup_{n \rightarrow \infty} |a_n|^{1/n}, \quad R = \frac{1}{L}$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$. Then the power series converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$.

Proof. Apply the root test to $a_n(z - z_0)^n$. The n th root of its magnitude is

$$|a_n|^{1/n} |z - z_0|,$$

whose limsup is $L|z - z_0|$. It is < 1 exactly inside the radius and > 1 outside. $\square$

Deep intuition

The coefficients set a geometric scale. The variable contributes a factor $|z - z_0|^n$, while the coefficients contribute roughly L^n . The competition is between the exponential rates L and $1/|z - z_0|$. The boundary is where neither wins automatically.

Theorem 7.10 (Uniform convergence on smaller disks). *If a power series has radius $R > 0$, then for every $0 < r < R$ it converges absolutely and uniformly on $|z - z_0| \leq r$.*

Proof. Choose ρ with $r < \rho < R$. Since the series converges absolutely at any point with modulus ρ , the numerical series $\sum |a_n| \rho^n$ converges. For $|z - z_0| \leq r < \rho$,

$$|a_n(z - z_0)^n| \leq |a_n| \rho^n.$$

The M-test gives uniform absolute convergence. $\square$

Theorem 7.11 (Termwise differentiation). *Inside its radius of convergence, a power series may be differentiated term by term:*

$$\frac{d}{dz} \sum_{n=0}^{\infty} a_n(z - z_0)^n = \sum_{n=1}^{\infty} n a_n(z - z_0)^{n-1},$$

and the differentiated series has the same radius of convergence.

Proof. First, the new coefficients have the same exponential growth because $n^{1/n} \rightarrow 1$:

$$\limsup |n a_n|^{1/n} = \limsup n^{1/n} |a_n|^{1/n} = L.$$

So the derivative series has the same radius R . Fix $r < R$. On $|z - z_0| \leq r$, the derivative series converges uniformly by the preceding theorem. The original partial sums are polynomials, hence differentiable, and they converge at z_0 (indeed they all equal a_0 there). The uniform convergence of derivatives theorem therefore applies and identifies the derivative of the limit with the limit of the derivatives. $\square$

Example 7.12 (A diagnostic power series). For

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} z^n,$$

we have

$$\limsup \left(\frac{n^2}{3^n} \right)^{1/n} = \frac{1}{3},$$

so $R = 3$. On $|z| = 3$, the terms have magnitude n^2 and therefore do not tend to zero; the series diverges at every boundary point.

Why this matters later

Power series are the bridge from real analysis to complex rigidity. In complex analysis, holomorphic functions will turn out not merely differentiable but locally equal to convergent power series. That single fact drives isolated zeros, the identity theorem, and residue calculus.

Part V

Complex analysis: rigidity from complex differentiability

Complex differentiability and Cauchy theory

8.1 Why complex differentiability is unusually rigid

Definition 8.1. Let $U \subseteq \mathbb{C}$ be open. A function $f : U \rightarrow \mathbb{C}$ is *complex differentiable at z_0* if the limit

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists as a complex number. It is *holomorphic* on U if it is complex differentiable at every point of U .

Why this idea exists

The formula looks identical to real differentiation, but the input increment h can approach 0 from infinitely many directions in the plane. A single complex number must describe the first-order behavior in every direction simultaneously. That compatibility makes holomorphic functions far more rigid than ordinary differentiable functions of two real variables.

Deep intuition

Multiplication by a nonzero complex number $a = re^{i\theta}$ is exactly a rotation by θ followed by a scaling by r . Thus if f is complex differentiable at z_0 , its first-order behavior is

$$f(z_0 + h) = f(z_0) + f'(z_0)h + o(|h|).$$

Locally, to first order, f is a rotation-and-scaling, not an arbitrary linear distortion. The Cauchy–Riemann equations are the coordinate expression of this fact.

Example ladder

Holomorphic. $f(z) = z^2$ has

$$\frac{(z+h)^2 - z^2}{h} = 2z + h \rightarrow 2z.$$

Not holomorphic. $f(z) = \bar{z}$ gives quotient 1 along real h and -1 along purely imaginary h , so no complex derivative exists anywhere.

8.2 Cauchy–Riemann equations

Write $z = x + iy$ and

$$f(z) = u(x, y) + iv(x, y).$$

Theorem 8.2 (Cauchy–Riemann, necessary form). *If f is complex differentiable at $z_0 = (x_0, y_0)$ and the relevant real partial derivatives exist, then*

$$u_x = v_y, \quad u_y = -v_x$$

at z_0 .

Proof. Approach with real increments $h \in \mathbb{R}$:

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{u(x_0 + h, y_0) - u(x_0, y_0)}{h} + i \frac{v(x_0 + h, y_0) - v(x_0, y_0)}{h} = u_x + iv_x.$$

Approach with imaginary increments ih :

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{u(x_0, y_0 + h) - u(x_0, y_0) + i(v(x_0, y_0 + h) - v(x_0, y_0))}{ih} = v_y - iu_y.$$

Equating real and imaginary parts yields $u_x = v_y$ and $v_x = -u_y$. □

Theorem 8.3 (Cauchy–Riemann, sufficient form). *Suppose u and v have continuous first partial derivatives in a neighborhood of (x_0, y_0) and satisfy the Cauchy–Riemann equations there. Then $f = u + iv$ is holomorphic there, with*

$$f' = u_x + iv_x = v_y - iu_y.$$

Proof. Continuity of the first partial derivatives implies real differentiability. For $h = \Delta x + i\Delta y$,

$$\Delta u = u_x \Delta x + u_y \Delta y + o(|h|), \quad \Delta v = v_x \Delta x + v_y \Delta y + o(|h|),$$

where the derivatives are evaluated at the base point. Using $v_x = -u_y$ and $v_y = u_x$,

$$\Delta f = (u_x + iv_x)\Delta x + (u_y + iv_y)\Delta y + o(|h|) = (u_x + iv_x)(\Delta x + i\Delta y) + o(|h|).$$

Thus

$$\frac{\Delta f}{h} = u_x + iv_x + o(1),$$

so the complex derivative exists. □

Example 8.4 (Why $\bar{z}$ fails). For $f(z) = \bar{z} = x - iy$, $u = x$ and $v = -y$. Thus $u_x = 1$ while $v_y = -1$, so the Cauchy–Riemann equations fail everywhere.

8.3 Contours and complex integration

Definition 8.5. A piecewise C^1 path is a map $\gamma : [a, b] \rightarrow \mathbb{C}$ that is continuously differentiable on finitely many subintervals. For continuous f on the image of γ ,

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Deep intuition

The factor $\gamma'(t)$ converts parameter motion into an oriented complex displacement. A contour integral therefore accumulates values of f while remembering direction. Reversing the path reverses the sign.

Example 8.6 (A circle). For $\gamma(t) = Re^{it}$, $0 \leq t \leq 2\pi$,

$$\int_{|z|=R} \frac{dz}{z} = \int_0^{2\pi} \frac{iRe^{it}}{Re^{it}} dt = 2\pi i.$$

But for an integer $k \neq -1$,

$$\int_{|z|=R} z^k dz = iR^{k+1} \int_0^{2\pi} e^{i(k+1)t} dt = 0.$$

This is the first glimpse of why the coefficient of $(z - a)^{-1}$ will be special.

Proposition 8.7 (ML estimate). If $|f(z)| \leq M$ on a contour γ of length L , then

$$\left| \int_{\gamma} f(z) dz \right| \leq ML.$$

Proof. By the triangle inequality for the real integral,

$$\left| \int_a^b f(\gamma(t)) \gamma'(t) dt \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| dt \leq M \int_a^b |\gamma'(t)| dt = ML.$$

□

Proposition 8.8 (Fundamental theorem for contour integrals). If F is holomorphic on U and $F' = f$, then for every piecewise C^1 path γ from a to b in U ,

$$\int_{\gamma} f(z) dz = F(b) - F(a).$$

In particular, closed contour integrals vanish.

Proof. On each smooth piece, the real chain rule gives

$$\frac{d}{dt} F(\gamma(t)) = F'(\gamma(t)) \gamma'(t) = f(\gamma(t)) \gamma'(t).$$

Integrate and telescope across the pieces.

□

8.4 Goursat's theorem: holomorphicity creates local cancellation

Theorem 8.9 (Goursat triangle theorem). *If f is holomorphic on an open set containing a closed triangle T and its interior, then*

$$\int_{\partial T} f(z) dz = 0.$$

Proof. Let $I(T) = \int_{\partial T} f(z) dz$. Join the midpoints of the sides, subdividing T into four congruent triangles. The integrals over internal edges cancel, so $I(T)$ is the sum of the four subtriangle boundary integrals. Hence one subtriangle T_1 satisfies

$$|I(T_1)| \geq \frac{1}{4} |I(T)|.$$

Repeat to obtain nested triangles $T \supset T_1 \supset T_2 \supset \cdots$ whose diameters are $d_n = 2^{-n}d_0$ and such that

$$|I(T_n)| \geq 4^{-n} |I(T)|.$$

Their intersection is a single point z_0 . Differentiability at z_0 gives

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + r(z)(z - z_0), \quad r(z) \rightarrow 0.$$

The first two terms have primitives, so their integrals around ∂T_n vanish. Therefore

$$I(T_n) = \int_{\partial T_n} r(z)(z - z_0) dz.$$

Let $\eta_n = \sup_{z \in T_n} |r(z)| \rightarrow 0$. The perimeter of T_n is at most Cd_n for a fixed C , and $|z - z_0| \leq d_n$, so the ML estimate yields

$$|I(T_n)| \leq C\eta_n d_n^2 = C'\eta_n 4^{-n}.$$

Combining with the lower bound gives $|I(T)| \leq C'\eta_n \rightarrow 0$. Hence $I(T) = 0$. $\square$

Deep intuition

The proof repeatedly zooms into the worst possible triangle. Differentiability says that at sufficiently small scale, f is nearly affine. Affine functions have primitives, so their loop integrals vanish. The remaining nonlinear error becomes too small to support a nonzero integral.

8.5 Cauchy's theorem on star-shaped domains

Definition 8.10. An open set U is *star-shaped with center a* if the line segment from a to every $z \in U$ lies entirely in U .

Theorem 8.11 (Cauchy theorem, star-shaped form). *If U is star-shaped and f is holomorphic on U , then f has a primitive on U . Consequently every closed piecewise C^1 contour in U has integral 0.*

Proof. Fix a star center a and define

$$F(z) = \int_{[a,z]} f(w) dw.$$

For small h , the triangle with vertices $a, z, z + h$ lies in U . Goursat's theorem gives

$$\int_{[a,z]} f + \int_{[z,z+h]} f + \int_{[z+h,a]} f = 0,$$

so

$$F(z+h) - F(z) = \int_{[z,z+h]} f(w) dw.$$

Parametrize $w = z + th$:

$$\frac{F(z+h) - F(z)}{h} = \int_0^1 f(z+th) dt \rightarrow f(z)$$

by continuity. Hence $F' = f$, and closed integrals vanish by the fundamental theorem for contour integrals. $\square$

Deep intuition

Holomorphic functions have local antiderivatives because tiny loops cancel. A star-shaped domain lets us choose paths from one center to every point without ambiguity. That converts local cancellation into a global primitive.

8.6 Homotopy invariance: when loops can be deformed

For this analytic refresher we use the following explicitly smooth notion, avoiding any hidden appeal to a smoothing theorem from topology.

Definition 8.12. Two closed piecewise C^1 curves $\gamma_0, \gamma_1 : [0, 1] \rightarrow U$ are *smoothly homotopic through closed curves* if there is a piecewise C^2 map

$$H : [0, 1]^2 \rightarrow U$$

with

$$H(0, t) = \gamma_0(t), \quad H(1, t) = \gamma_1(t), \quad H(s, 0) = H(s, 1).$$

We call U *smoothly simply connected* if every closed piecewise C^1 curve is smoothly homotopic to a constant curve.

Theorem 8.13 (Homotopy invariance). *If f is holomorphic on U and γ_0, γ_1 are smoothly homotopic through closed curves, then*

$$\int_{\gamma_0} f(z) dz = \int_{\gamma_1} f(z) dz.$$

Proof. First assume H is C^2 on the whole square; subdividing handles the piecewise case. Define

$$I(s) = \int_0^1 f(H(s, t)) H_t(s, t) dt.$$

Continuity on the compact square justifies differentiation under the integral. The chain and product rules give

$$I'(s) = \int_0^1 [f'(H) H_s H_t + f(H) H_{st}] dt = \int_0^1 \frac{\partial}{\partial t} (f(H) H_s) dt.$$

Hence

$$I'(s) = f(H(s, 1)) H_s(s, 1) - f(H(s, 0)) H_s(s, 0).$$

Since $H(s, 1) = H(s, 0)$ for every s , differentiating this identity gives $H_s(s, 1) = H_s(s, 0)$. Therefore $I'(s) = 0$, so I is constant and the endpoint contour integrals agree. $\square$

Corollary 8.14 (Cauchy theorem, smoothly simply connected form). *If U is smoothly simply connected and f is holomorphic on U , then every closed piecewise C^1 contour in U has integral 0.*

Proof. Deform the contour to a constant contour. Homotopy invariance preserves the integral, and a constant contour has derivative 0 and hence integral 0. $\square$

Common pitfall

The slogan “holomorphic closed integrals are zero” needs a domain hypothesis. On $\mathbb{C} \setminus \{0\}$, the function $1/z$ is holomorphic, but

$$\int_{|z|=1} \frac{dz}{z} = 2\pi i.$$

The puncture prevents the circle from shrinking to a point without crossing the missing point.

8.7 Cauchy’s integral formula: the boundary determines the interior

Lemma 8.15. *If C is the positively oriented circle $|z - a| = R$ and z_0 lies inside it, then*

$$\int_C \frac{dz}{z - z_0} = 2\pi i.$$

Proof. Write $z = a + Re^{it}$ and $w = (z_0 - a)/R$, so $|w| < 1$. Then

$$\int_C \frac{dz}{z - z_0} = i \int_0^{2\pi} \frac{dt}{1 - we^{-it}}.$$

The geometric series $\sum_{n \geq 0} w^n e^{-int}$ converges uniformly in t , so integrate termwise:

$$i \sum_{n \geq 0} w^n \int_0^{2\pi} e^{-int} dt = 2\pi i.$$

Only the $n = 0$ term survives. $\square$

Theorem 8.16 (Cauchy integral formula). *Let f be holomorphic on an open set containing the closed disk $\overline{D}(a, R)$. If $z_0 \in D(a, R)$ and C is its positively oriented boundary circle, then*

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz.$$

Proof. Define

$$g(z) = \begin{cases} \frac{f(z) - f(z_0)}{z - z_0}, & z \neq z_0, \\ f'(z_0), & z = z_0. \end{cases}$$

The removable definition makes g holomorphic on the disk. Cauchy's theorem gives $\int_C g(z) dz = 0$. Therefore

$$\int_C \frac{f(z)}{z - z_0} dz = f(z_0) \int_C \frac{dz}{z - z_0} = 2\pi i f(z_0).$$

□

Deep intuition

This is the first major rigidity theorem. A holomorphic function cannot choose its interior values independently of its boundary values. One entire boundary circle contains enough information to reconstruct the value at every interior point.

8.8 Derivatives, Cauchy estimates, and rigidity

Theorem 8.17 (Derivative form of Cauchy's formula). *Under the same hypotheses,*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Proof. Fix a compact smaller disk inside C . The kernel $(z - w)^{-1}$ and all its derivatives with respect to w are uniformly bounded for $z \in C$ and w in the smaller disk. Therefore differentiation under the integral is justified. Repeated differentiation of

$$\frac{1}{z - w}$$

produces $n!/(z - w)^{n+1}$, yielding the formula. □

Theorem 8.18 (Cauchy estimates). *If f is holomorphic on a neighborhood of $\overline{D}(z_0, R)$ and $|f(z)| \leq M$ on $|z - z_0| = R$, then*

$$|f^{(n)}(z_0)| \leq \frac{n!M}{R^n}.$$

Proof. Apply the derivative formula and the ML estimate. The circle has length $2\pi R$ and $|z - z_0| = R$, so

$$|f^{(n)}(z_0)| \leq \frac{n!}{2\pi} \frac{M}{R^{n+1}} (2\pi R) = \frac{n!M}{R^n}.$$

□

Theorem 8.19 (Liouville). *Every bounded entire function is constant.*

Proof. Suppose $|f(z)| \leq M$ on all of $\mathbb{C}$. For any $R > 0$, Cauchy's estimate at z_0 gives

$$|f'(z_0)| \leq \frac{M}{R}.$$

Let $R \rightarrow \infty$. Then $f'(z_0) = 0$. Since z_0 was arbitrary, $f' \equiv 0$, so f is constant. $\square$

Theorem 8.20 (Fundamental theorem of algebra). *Every nonconstant polynomial with complex coefficients has a complex root.*

Proof. Suppose a nonconstant polynomial p has no zero. Then $1/p$ is entire. As $|z| \rightarrow \infty$, the leading term dominates, so $|p(z)| \rightarrow \infty$ and hence $1/p(z) \rightarrow 0$. Therefore $1/p$ is bounded outside a large disk; it is also bounded on the compact disk by continuity. Liouville's theorem makes $1/p$ constant, impossible because p is nonconstant. $\square$

Deep intuition

Cauchy's formula converts distance to the boundary into derivative control. If an entire function is bounded, you can place the boundary arbitrarily far away while keeping the same bound, forcing every derivative to vanish. This is the basic rigidity mechanism: analytic information at large scales crushes local freedom.

Power series, zeros, Laurent expansions, and residues

9.1 Holomorphic functions are analytic

Theorem 9.1 (Taylor expansion). *If f is holomorphic on a neighborhood of $\overline{D(a, R)}$, then for $|z - a| < R$,*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n,$$

and the series converges uniformly on every smaller closed disk.

Proof. Choose r with $|z - a| < r < R$ and apply Cauchy's formula on $|\zeta - a| = r$:

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Because $|z - a| < r = |\zeta - a|$,

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - a} \frac{1}{1 - (z - a)/(\zeta - a)} = \sum_{n=0}^{\infty} \frac{(z - a)^n}{(\zeta - a)^{n+1}}.$$

The geometric series converges uniformly on the contour, so integrate termwise:

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta \right) (z - a)^n.$$

By the derivative form of Cauchy's formula, the coefficient is $f^{(n)}(a)/n!$. Uniform convergence on smaller disks follows from geometric uniformity or from the power-series theorem of Chapter 7. $\square$

Deep intuition

Real differentiability gives local linear approximation. Complex differentiability, after Cauchy's theorem, recursively gives all derivatives and an actual convergent power series. A holomorphic function carries an infinite amount of local structure.

9.2 Zeros, local factorization, and the identity theorem

Proposition 9.2 (Local zero dichotomy). *Let f be holomorphic near a and suppose $f(a) = 0$. Exactly one of the following happens:*

1. *all Taylor coefficients of f at a vanish, in which case f is identically zero on some neighborhood of a ;*
2. *there is a unique integer $m \geq 1$ and a holomorphic g with $g(a) \neq 0$ such that*

$$f(z) = (z - a)^m g(z).$$

In the second case, a is an isolated zero.

Proof. Expand f in its Taylor series about a . If every coefficient is zero, the series is identically zero on its disk of convergence, giving the first case. Otherwise let m be the least index with nonzero coefficient c_m . Then

$$f(z) = (z - a)^m (c_m + c_{m+1}(z - a) + \cdots).$$

The parenthesized series defines a holomorphic g with $g(a) = c_m \neq 0$. Continuity makes g nonzero on a small neighborhood, so the only zero there is a . Uniqueness of m follows from the least nonzero Taylor coefficient. $\square$

Theorem 9.3 (Identity theorem). *If holomorphic functions f, g on a connected open set U agree on a set having an accumulation point in U , then $f \equiv g$ on U .*

Proof. Let $h = f - g$. Let

$$A = \{z \in U : h \text{ vanishes on some neighborhood of } z\}.$$

The set A is open by definition. The assumed accumulation point a of zeros belongs to A : if the second alternative of the local zero dichotomy held at a , then a would be an isolated zero, contradicting the existence of other zeros arbitrarily close to a . Hence h vanishes near a and A is nonempty.

We show A is closed relative to U . Suppose $a_n \in A$ and $a_n \rightarrow a \in U$. Then $h(a_n) = 0$ for all n . If infinitely many a_n are distinct from a , they make a an accumulation point of zeros, so the same local dichotomy implies that h vanishes near a , hence $a \in A$. If the sequence is eventually equal to a , then $a \in A$ already. Thus A is closed.

Since U is connected and A is nonempty, open, and closed in U , we have $A = U$. Therefore $h \equiv 0$, i.e. $f \equiv g$. $\square$

Corollary 9.4 (Zeros are isolated). *If U is connected and f is holomorphic on U but not identically zero, then every zero of f is isolated.*

Proof. At a zero a , the first alternative of the local zero dichotomy would make f vanish on a neighborhood. The identity theorem would then force $f \equiv 0$ on U , contrary to hypothesis. Hence the second alternative holds and the zero is isolated. $\square$

Deep intuition

For an ordinary smooth function, a zero set can be wildly complicated. For a nonzero holomorphic function, a zero has a first nonzero Taylor term, and that term behaves like $(z - a)^m$. The zero is therefore a discrete event with a well-defined multiplicity. If zeros ever begin to accumulate, the only way out is that every Taylor coefficient vanishes and the function becomes identically zero.

9.3 Laurent series: Taylor series with a controlled hole

Definition 9.5. A Laurent series about a is a doubly infinite series

$$\sum_{n=-\infty}^{\infty} c_n(z - a)^n.$$

Negative powers allow singular behavior at a while retaining power-series control away from a .

Why this idea exists

Taylor series describe holomorphic behavior in a disk. If the center itself is missing, a Taylor series is too restrictive: $1/(z - a)$ is perfectly holomorphic on the punctured disk but has no Taylor series there. Laurent series are the correct coordinate system for a neighborhood with one hole.

Lemma 9.6 (Annular Cauchy representation). *Suppose f is holomorphic on an annulus*

$$r < |\zeta - a| < R.$$

Choose radii $r < \rho_1 < |z - a| < \rho_2 < R$. Then

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta-a|=\rho_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{|\zeta-a|=\rho_1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Proof. Cut the closed annular region $\rho_1 \leq |\zeta - a| \leq \rho_2$ along a radial segment not passing through z , and remove a small disk around z . The remaining compact region can be subdivided into finitely many small curvilinear rectangles, each lying in a disk where the Cauchy theorem applies; equivalently, approximate the circular boundaries by polygonal ones and triangulate. Summing the zero boundary integrals cancels all interior edges. The two sides of the radial cut occur with opposite orientations and cancel. The resulting boundary identity is

$$\int_{|\zeta-a|=\rho_2} \frac{f(\zeta)}{\zeta - z} d\zeta - \int_{|\zeta-a|=\rho_1} \frac{f(\zeta)}{\zeta - z} d\zeta - \int_{|\zeta-z|=\delta} \frac{f(\zeta)}{\zeta - z} d\zeta = 0.$$

As $\delta \rightarrow 0$, write $f(\zeta) = f(z) + o(1)$ on the small circle. Its integral tends to $2\pi i f(z)$, because the error is bounded by $o(1) \cdot 2\pi$ and $\int d\zeta/(\zeta - z) = 2\pi i$. Rearranging gives the formula. $\square$

Theorem 9.7 (Laurent expansion). *If f is holomorphic on the annulus $r < |z - a| < R$, then there is a unique Laurent series*

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z - a)^n$$

converging absolutely and uniformly on every compact subannulus. The coefficients are

$$c_n = \frac{1}{2\pi i} \int_{|\zeta-a|=\rho} \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta,$$

for any ρ with $r < \rho < R$.

Proof. Fix z and radii $\rho_1 < |z-a| < \rho_2$. For the outer integral in the annular representation,

$$\frac{1}{\zeta-z} = \frac{1}{\zeta-a} \sum_{n=0}^{\infty} \left(\frac{z-a}{\zeta-a} \right)^n,$$

which converges uniformly on $|\zeta-a| = \rho_2$. This gives nonnegative powers

$$\sum_{n=0}^{\infty} c_n (z-a)^n.$$

For the inner integral, since $|\zeta-a| < |z-a|$,

$$\frac{1}{\zeta-z} = -\frac{1}{z-a} \frac{1}{1 - (\zeta-a)/(z-a)} = -\sum_{m=0}^{\infty} \frac{(\zeta-a)^m}{(z-a)^{m+1}}.$$

The minus sign in the annular formula turns this into the negative-power part. Uniform geometric convergence on compact subannuli justifies termwise integration and gives uniform convergence there.

The coefficient formula follows from integrating the Laurent series termwise on a circle: all powers integrate to zero except the exponent -1 . This also proves uniqueness. Independence of the chosen radius follows because the coefficient is uniquely determined by the function. $\square$

Deep intuition

A Laurent series decomposes behavior into angular modes. Around a circle, $(z-a)^n$ winds n times. Under contour integration, every mode cancels over one revolution except $(z-a)^{-1}$, whose phase is exactly canceled by dz . The residue is the one mode that survives a loop.

9.4 Classifying isolated singularities

If f is holomorphic on $0 < |z-a| < R$ with Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n,$$

then:

- a is *removable* if all negative coefficients vanish;
- a is a *pole of order m* if $c_{-m} \neq 0$ and $c_n = 0$ for all $n < -m$;

- a is an *essential singularity* if infinitely many negative coefficients are nonzero.

Theorem 9.8 (Riemann removable singularity criterion). *If f is holomorphic on $0 < |z - a| < R$ and bounded near a , then a is removable.*

Proof. For $n \geq 1$, the negative Laurent coefficient is

$$c_{-n} = \frac{1}{2\pi i} \int_{|\zeta - a| = \rho} f(\zeta)(\zeta - a)^{n-1} d\zeta.$$

If $|f| \leq M$, the ML estimate gives

$$|c_{-n}| \leq \frac{1}{2\pi} M \rho^{n-1} (2\pi \rho) = M \rho^n.$$

Let $\rho \rightarrow 0$ to obtain $c_{-n} = 0$. Thus the principal part vanishes and the Laurent series is an ordinary Taylor series extending across a . $\square$

Example ladder

$\frac{e^z - 1}{z}$ has a removable singularity at 0 because $e^z - 1 = z + z^2/2! + \dots$, so the quotient extends with value 1.

$1/z^3$ has a pole of order 3 at 0.

$e^{1/z} = 1 + z^{-1} + \frac{1}{2!}z^{-2} + \dots$ has an essential singularity at 0 because infinitely many negative powers occur.

9.5 Residues: the coefficient that a loop can detect

Definition 9.9. The *residue* of f at an isolated singularity a is the coefficient c_{-1} in its Laurent expansion:

$$\text{Res}(f; a) = c_{-1}.$$

Equivalently,

$$\text{Res}(f; a) = \frac{1}{2\pi i} \int_{|z - a| = \rho} f(z) dz$$

for sufficiently small ρ .

Deep intuition

Residue is not an arbitrary coefficient singled out by convention. It is the unique local coefficient that survives integration around a loop. All other Laurent modes cancel over one revolution.

Proposition 9.10 (Simple pole formula). *If $f = g/h$ with g, h holomorphic near a , $h(a) = 0$, $h'(a) \neq 0$, and $g(a)$ finite, then*

$$\text{Res}(f; a) = \frac{g(a)}{h'(a)}.$$

Proof. Since h has a simple zero, write $h(z) = (z-a)k(z)$ with k holomorphic and $k(a) = h'(a) \neq 0$. Then

$$f(z) = \frac{1}{z-a} \frac{g(z)}{k(z)},$$

and the second factor is holomorphic with value $g(a)/h'(a)$. Therefore that value is the coefficient of $(z-a)^{-1}$. $\square$

Theorem 9.11 (Residue theorem for a circle). *Let f be meromorphic on a neighborhood of a closed disk and suppose its poles $a_1, \dots, a_m$ lie strictly inside the boundary circle C . Then*

$$\int_C f(z) dz = 2\pi i \sum_{j=1}^m \text{Res}(f; a_j).$$

Proof. At each pole a_j , let $P_j(z)$ be the finite principal part of the Laurent expansion. Then

$$h(z) = f(z) - \sum_{j=1}^m P_j(z)$$

has removable singularities at every a_j and therefore extends holomorphically to the whole disk. Cauchy's theorem gives $\int_C h = 0$. Each principal part is a finite sum of terms $c_{-k}(z-a_j)^{-k}$. For $k \neq 1$, the contour integral is zero; for $k = 1$, it is $2\pi i c_{-1}$ by Cauchy's integral formula applied to the constant function 1. Summing gives the result. $\square$

Example 9.12 (A three-pole integral). Evaluate

$$\int_{|z|=3} \frac{z^2 + 1}{z(z-1)(z-2)} dz.$$

All three simple poles 0, 1, 2 lie inside. Their residues are

$$\text{Res}(f; 0) = \frac{1}{(-1)(-2)} = \frac{1}{2},$$

$$\text{Res}(f; 1) = \frac{2}{1(1-2)} = -2,$$

$$\text{Res}(f; 2) = \frac{5}{2(2-1)} = \frac{5}{2}.$$

The sum is 1, so the integral equals $2\pi i$.

9.6 Logarithmic derivatives: zeros and poles become additive

Proposition 9.13. *If f has a zero of order m at a , then*

$$\text{Res}\left(\frac{f'}{f}; a\right) = m.$$

If f has a pole of order m , the residue is $-m$.

Proof. For a zero, write $f(z) = (z - a)^m g(z)$ with $g(a) \neq 0$. Then

$$\frac{f'}{f} = \frac{m}{z - a} + \frac{g'}{g},$$

and g'/g is holomorphic near a . Thus the residue is m . For a pole, write $f = (z - a)^{-m} g$ and obtain residue $-m$. $\square$

Deep intuition

The logarithmic derivative turns multiplication into addition:

$$\frac{(fg)'}{fg} = \frac{f'}{f} + \frac{g'}{g}.$$

A product of many local factors becomes a sum of simple reciprocal terms. This is exactly why logarithmic derivatives are so effective in number theory: Euler products over primes become prime-weighted sums.

Theorem 9.14 (Argument principle). *Let f be meromorphic on a neighborhood of a closed disk, with no zero or pole on its boundary C . If N is the number of zeros inside, counted with multiplicity, and P the number of poles, counted with order, then*

$$\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz = N - P.$$

Proof. The poles of f'/f occur exactly at zeros and poles of f . By the preceding proposition, the residues are the corresponding positive zero multiplicities and negative pole orders. Apply the residue theorem and sum them. $\square$

9.7 Uniform limits of holomorphic functions

Theorem 9.15 (Weierstrass theorem). *If f_n are holomorphic on an open set U and converge uniformly on every compact subset of U to f , then f is holomorphic. Moreover $f_n^{(k)} \rightarrow f^{(k)}$ uniformly on compact subsets for every k .*

Proof. Fix a closed disk $\overline{D(a, R)} \subset U$ and a smaller disk $\overline{D(a, r)}$ with $r < R$. Uniform convergence on the boundary circle gives, by Cauchy's formula,

$$f_n(z) = \frac{1}{2\pi i} \int_{|\zeta - a| = R} \frac{f_n(\zeta)}{\zeta - z} d\zeta.$$

Let $n \rightarrow \infty$. Uniform convergence allows the limit through the integral, yielding the same formula for f . Differentiating the integral kernel shows f is holomorphic and

$$f_n^{(k)}(z) - f^{(k)}(z) = \frac{k!}{2\pi i} \int_{|\zeta - a| = R} \frac{f_n(\zeta) - f(\zeta)}{(\zeta - z)^{k+1}} d\zeta.$$

For $|z - a| \leq r$, the denominator is at least $(R - r)^{k+1}$, so the right-hand side tends to zero uniformly on the smaller disk. Cover compact subsets by finitely many such disks. $\square$

Why this matters later

This theorem is one reason analytic number theory can build new holomorphic functions from infinite sums and products: once convergence is controlled locally uniformly, holomorphic structure survives the limit.

Part VI

Synthesis: why the subjects meet

From roots and primes to Frobenius and Euler products

10.1 Roots of unity: algebra sitting inside complex analysis

For $n \geq 1$, define

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\}.$$

Theorem 10.1. *The group μ_n is cyclic of order n , generated by*

$$\zeta_n = e^{2\pi i/n}.$$

Moreover ζ_n^a is a generator if and only if $\gcd(a, n) = 1$.

Proof. The solutions of $z^n = 1$ are exactly

$$e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1,$$

so $\mu_n = \langle \zeta_n \rangle$ and has n elements. The order of ζ_n^a is the least positive m with $n \mid am$, namely $n/\gcd(a, n)$. This equals n exactly when $\gcd(a, n) = 1$. $\square$

Deep intuition

Roots of unity are simultaneously geometric points on a circle, algebraic roots of a polynomial, and elements of a finite group. Cyclotomic fields exploit all three identities at once. They are an ideal first example of how algebraic number theory refuses to keep geometry, algebra, and arithmetic separate.

10.2 Factorization over finite fields is Frobenius cycle structure

Let $f \in \mathbb{F}_q[x]$ be square-free and split over some finite extension. Frobenius $F : x \mapsto x^q$ permutes the roots.

Theorem 10.2 (Factor degrees equal Frobenius cycle lengths). *Suppose*

$$f = f_1 \cdots f_r$$

is the factorization of a square-free polynomial into distinct monic irreducibles over $\mathbb{F}_q$, with $\deg f_i = d_i$. Then Frobenius acting on the roots of f has disjoint cycle lengths

$$d_1, \dots, d_r.$$

Proof. For each irreducible factor f_i , choose a root α_i . By the finite-field orbit theorem, the roots of f_i are exactly the Frobenius orbit

$$\alpha_i, \alpha_i^q, \dots, \alpha_i^{q^{d_i}-1},$$

which is a cycle of length d_i . Distinct irreducible factors have disjoint root sets because f is square-free. Hence the full Frobenius permutation is the disjoint union of these cycles. $\square$

Example 10.3 (An irreducible cubic over $\mathbb{F}_2$). The polynomial $x^3 + x + 1$ has no root in $\mathbb{F}_2$, so as a cubic it is irreducible. If α is a root in $\mathbb{F}_8$, the roots are

$$\alpha, \alpha^2, \alpha^4,$$

and the next Frobenius iterate is $\alpha^8 = \alpha$. Thus Frobenius acts as a 3-cycle.

Deep intuition

Factoring a polynomial modulo a prime is not merely computational simplification. The factor degrees are a compressed record of how Frobenius permutes the roots. Arithmetic factorization is reading off a group action.

10.3 A concrete S_3 preview: $x^3 - 2$ modulo primes

Return to $f(x) = x^3 - 2$, whose splitting field over $\mathbb{Q}$ has Galois group S_3 . Away from the bad primes 2 and 3 (the primes dividing the discriminant), reduction is square-free. Three examples show all three conjugacy classes of S_3 .

Example ladder

Modulo 5: factor type 1 + 2.

$$x^3 - 2 \equiv (x + 2)(x^2 - 2x - 1) \pmod{5},$$

and the quadratic is irreducible modulo 5. The corresponding Frobenius cycle type is (1)(2), a transposition.

Modulo 7: factor type 3. The cubes modulo 7 are 0, 1, 6, so $x^3 = 2$ has no solution. A cubic without a root is irreducible, so the cycle type is a 3-cycle.

Modulo 31: factor type 1 + 1 + 1.

$$x^3 - 2 \equiv (x - 4)(x - 7)(x + 11) \pmod{31}.$$

All roots lie in the base field, so Frobenius fixes each root: the cycle type is the identity.

Deep intuition

Chebotarev will eventually say, roughly, that as primes vary, these Frobenius conjugacy classes appear with frequencies proportional to the sizes of the conjugacy classes. For S_3 , the class sizes are 1 (identity), 3 (transpositions), and 2 (3-cycles), suggesting densities $1/6$, $1/2$, and $1/3$ after the finitely many bad primes are discarded. Phase 0 is about understanding every object in that sentence before proving the theorem.

10.4 Unique factorization becomes an Euler product

For complex s with $\Re(s) > 1$, define

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad n^{-s} = e^{-s \log n}.$$

Absolute convergence follows because $|n^{-s}| = n^{-\Re(s)}$ and the corresponding p -series converges.

Theorem 10.4 (Euler product). *For $\Re(s) > 1$,*

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}},$$

meaning that the finite products over primes $p \leq y$ converge to $\zeta(s)$ as $y \rightarrow \infty$.

Proof. For a finite set P of primes,

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \prod_{p \in P} (1 + p^{-s} + p^{-2s} + \cdots).$$

The finite product of absolutely convergent geometric series may be expanded. Each choice of exponents $(e_p)_{p \in P}$ contributes

$$\prod_{p \in P} p^{-e_p s} = n^{-s},$$

where n has no prime factors outside P . Unique factorization says every such n occurs exactly once. Thus

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ p|n \Rightarrow p \in P}} n^{-s}.$$

As P increases through the primes, the right-hand side tends to $\zeta(s)$ by absolute convergence. This yields the product identity. Absolute convergence of the product can be seen from

$$\sum_p |p^{-s}| \leq \sum_{n=2}^{\infty} n^{-\Re(s)} < \infty,$$

which controls the product through the convergent expansion of $-\log(1 - z)$ for small z . $\square$

Deep intuition

The Dirichlet series $\sum n^{-s}$ lists integers additively, one by one. The Euler product reorganizes the same information multiplicatively, prime by prime. Unique factorization is exactly the dictionary that makes the two encodings identical.

Example 10.5 (Finite Euler product). With only primes 2 and 3,

$$(1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1} = \sum_{a,b \geq 0} (2^a 3^b)^{-s}.$$

The integers appearing are precisely those whose prime factors belong to $\{2, 3\}$: $1, 2, 3, 4, 6, 8, 9, 12, \dots$. This finite identity is the Euler product without any limiting subtlety.

10.5 Logarithmic derivatives turn prime products into prime sums

Theorem 10.6. For $\Re(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} (\log p) p^{-ks} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

where

$$\Lambda(n) = \begin{cases} \log p, & n = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 0, & \text{otherwise} \end{cases}$$

is the von Mangoldt function.

Proof. Fix $\delta > 0$ and work on $\Re(s) \geq 1 + \delta$. The double series

$$G(s) = \sum_p \sum_{k=1}^{\infty} \frac{p^{-ks}}{k}$$

converges locally uniformly, since

$$\sum_p \sum_{k \geq 1} \frac{|p^{-ks}|}{k} \leq \sum_p \sum_{k \geq 1} p^{-k(1+\delta)} \leq C_\delta \sum_p p^{-(1+\delta)} < \infty.$$

For finite sets of primes, the power-series identity $-\log(1 - w) = \sum_{k \geq 1} w^k/k$ gives

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \exp \left(\sum_{p \in P} \sum_{k \geq 1} \frac{p^{-ks}}{k} \right).$$

Passing to the limit and using the Euler product yields $\zeta(s) = e^{G(s)}$. Hence ζ is nonzero in this half-plane and

$$\frac{\zeta'(s)}{\zeta(s)} = G'(s).$$

The differentiated double series also converges locally uniformly because the extra factor $\log p$ is dominated by a small power of p . Therefore

$$G'(s) = - \sum_p \sum_{k=1}^{\infty} (\log p) p^{-ks}.$$

Every prime power $n = p^k$ appears exactly once and receives weight $\log p$, giving the von Mangoldt series. Since $\delta > 0$ was arbitrary, the identity holds throughout $\Re(s) > 1$. $\square$

Deep intuition

Products are the natural language of independent prime factors, but sums are easier for analysis. The logarithm turns products into sums, and differentiation removes the harmless $1/k$ weights. The logarithmic derivative is therefore a machine that converts multiplicative prime structure into an additive signal that analytic tools can measure.

10.6 Why conjugacy classes are the canonical output

Proposition 10.7. *Suppose a group G acts transitively on a set X , and to each $x \in X$ is attached an element $\sigma_x \in G$ satisfying*

$$\sigma_{g \cdot x} = g \sigma_x g^{-1}.$$

Then the conjugacy class of σ_x is independent of x .

Proof. If $y \in X$, transitivity gives $y = g \cdot x$ for some g . Then

$$\sigma_y = g \sigma_x g^{-1},$$

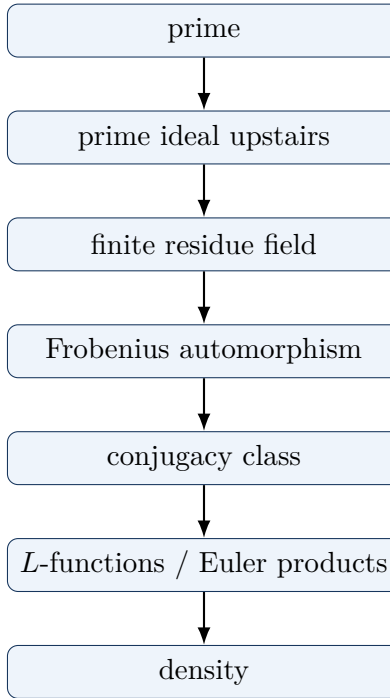
so σ_y and σ_x lie in the same conjugacy class. $\square$

Deep intuition

An individual Frobenius element is like a matrix representing a linear map after choosing a basis. Change the auxiliary choice and the matrix changes by conjugation. The conjugacy class is what remains after forgetting coordinates.

10.7 The emerging Chebotarev chain

The architecture to keep in mind is



Each arrow is now motivated by material already studied:

- ideals and quotients explain residue rings and fields;
- finite fields supply canonical Frobenius automorphisms;
- Galois theory turns root permutations into group elements;
- conjugation explains why only a class is canonical;
- Euler products encode primes analytically;
- logarithmic derivatives convert those products into sums that can be estimated.

Why this matters later

The next phase of study adds the genuinely arithmetic objects omitted here: rings of integers in number fields, prime ideal factorization, ramification, decomposition and inertia groups, Frobenius elements for unramified primes, and then the analytic machinery needed to prove a density theorem. The purpose of this reboot is that none of the algebraic or analytic language in that sentence should feel like an unrelated prerequisite.

Concept laboratories

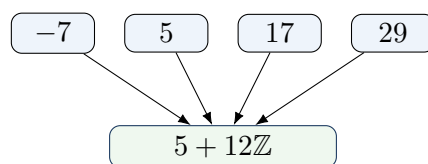
The purpose of these laboratories is not to introduce much new theory. It is to hold the same concept still long enough to see it from several angles. Each laboratory begins with a simple model, then pushes it until the structural reason for the definition becomes visible.

11.1 Laboratory I: quotient objects as deliberate loss of resolution

Start with the map

$$\pi : \mathbb{Z} \rightarrow \mathbb{Z}/12\mathbb{Z}.$$

At full resolution, 5, 17, and -7 are different integers. At resolution “modulo 12,” they are indistinguishable. The quotient is therefore not primarily a smaller set; it is a new standard of sameness.



The kernel $12\mathbb{Z}$ is exactly the collection of changes that the quotient refuses to notice. This gives a useful universal slogan:

Deep intuition

A kernel is a set of *invisible moves*. A quotient is the world obtained after declaring those moves irrelevant.

The same picture works for the exponential map

$$\mathbb{R} \rightarrow S^1, \quad t \mapsto e^{2\pi it}.$$

Moving by an integer is invisible, so the circle is $\mathbb{R}/\mathbb{Z}$. Geometry appears by quotienting a line by a periodicity.

Why does normality enter for groups? If N is not normal, the statement “ g and gn are the same” is not stable under multiplication from both sides. Changing a representative can leak back into

visible information. Normality says that invisible errors remain invisible after they are transported through any group element:

$$gng^{-1} \in N.$$

This is why normality is best remembered not as a conjugation formula but as *compatibility of forgetting with multiplication*.

Example 11.1 (A quotient failure in S_3). Let $H = \{e, (12)\}$. It is not normal because

$$(13)(12)(13)^{-1} = (23) \notin H.$$

Suppose left cosets could be multiplied by $(gH)(kH) = gkH$. Since $eH = (12)H$, replacing the first representative e by (12) should never change a product. But multiplying on the right by $(13)H$ would require

$$(13)H = (12)(13)H,$$

which is false. The representative choice becomes observable. That failure is exactly what normality prevents.

11.2 Laboratory II: one polynomial, several field enlargements

Consider

$$x^4 - 2.$$

This single polynomial lets us separate three questions that are easy to blur together.

Question 1: can we get one root? Let $\alpha = \sqrt[4]{2}$. Eisenstein at 2 shows $x^4 - 2$ is irreducible, so

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4.$$

This field solves the equation once, but it is real and therefore does not contain $i\alpha$ or $-i\alpha$.

Question 2: can we get all roots? The roots are

$$\alpha, -\alpha, i\alpha, -i\alpha.$$

The splitting field is

$$L = \mathbb{Q}(\alpha, i).$$

Since $\mathbb{Q}(\alpha) \subset \mathbb{R}$, $i \notin \mathbb{Q}(\alpha)$, and i has quadratic minimal polynomial over the real field. Hence

$$[L : \mathbb{Q}] = 8.$$

Question 3: what are the symmetries of the complete root picture? Define

$$r(\alpha) = i\alpha, \quad r(i) = i,$$

and let c be complex conjugation:

$$c(\alpha) = \alpha, \quad c(i) = -i.$$

Then r has order 4, c has order 2, and

$$crc = r^{-1}.$$

The eight maps r^k and cr^k are distinct automorphisms. Since $[L : \mathbb{Q}] = 8$, they are all automorphisms, so

$$\text{Gal}(L/\mathbb{Q}) \cong D_4,$$

the symmetry group of a square.

Deep intuition

This is Galois theory in its most literal form. The four roots sit at the vertices of a square in the complex plane. The Galois group is not merely *like* a symmetry group: it really is the group of algebraically permitted permutations of that square.

The example also separates three field-theoretic notions:

- $\mathbb{Q}(\alpha)/\mathbb{Q}$ is algebraic and separable but not normal;
- the splitting field $L/\mathbb{Q}$ is normal and separable, hence Galois;
- automorphisms become plentiful only after the whole conjugate root configuration is present.

11.3 Laboratory III: pointwise convergence can hide a moving defect

Return to $f_n(x) = x^n$ on $[0, 1]$. This example is worth studying beyond the standard “not uniform” conclusion.

For every fixed $x < 1$, the sequence eventually becomes tiny. Yet for each n there are points extremely close to 1 where x^n is still large. The error has not disappeared; it has been compressed into a thinner region and pushed toward the boundary.

Three different operations react differently:

1. **Continuity of the limit fails.** Every f_n is continuous, but the pointwise limit jumps at 1. This shows pointwise convergence is too weak to preserve local shape.
2. **Uniform convergence fails.** The supremum error remains close to 1 because the bad region, however thin, still contains some point.
3. **The integrals still converge.** In fact

$$\int_0^1 x^n dx = \frac{1}{n+1} \rightarrow 0 = \int_0^1 f(x) dx,$$

where changing the value at the single point 1 does not affect the Riemann integral.

Deep intuition

Different analytic operations “see” different kinds of error. The supremum norm sees the worst point and ignores how small the bad region is. Integration sees total area and may forgive a tall error if its width shrinks fast enough. Differentiation can do the opposite: it may magnify a tiny-amplitude but rapidly oscillating error.

This is why convergence hypotheses are not interchangeable. Uniform convergence is a powerful all-purpose guarantee, but sometimes a weaker theorem tailored to the operation is possible. Later analysis develops exactly such refined tools.

11.4 Laboratory IV: residues as a contour filter

Consider

$$f(z) = \frac{1}{(z-1)(z-3)}.$$

The same function gives different integrals depending on which singularities the contour encloses.

On $|z| = 2$, only the pole at 1 lies inside. Its residue is

$$\text{Res}(f; 1) = \frac{1}{1-3} = -\frac{1}{2},$$

so

$$\int_{|z|=2} f(z) dz = -\pi i.$$

On $|z| = 4$, both poles lie inside. The residue at 3 is $1/(3-1) = 1/2$, so the residues cancel and the integral is 0.

The partial fraction decomposition makes the mechanism transparent:

$$f(z) = -\frac{1}{2} \frac{1}{z-1} + \frac{1}{2} \frac{1}{z-3}.$$

A contour integral acts like a binary filter: each term contributes $2\pi i$ times its coefficient if its pole lies inside, and contributes 0 if its pole lies outside.

Deep intuition

Residue calculus is local-to-global compression. Instead of tracking a function along every point of a contour, you replace the entire integral by a finite list of local coefficients at the singularities trapped inside.

This also explains why contour choice matters. Moving a contour without crossing a pole changes nothing; crossing a pole changes the integral by exactly $2\pi i$ times that pole's residue.

11.5 Laboratory V: Frobenius as a permutation you can actually see

Take

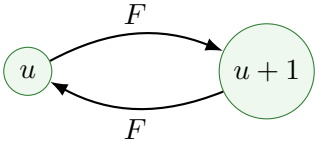
$$\mathbb{F}_4 = \{0, 1, u, u+1\}, \quad u^2 + u + 1 = 0.$$

The Frobenius map $x \mapsto x^2$ fixes 0 and 1 and swaps u with $u+1$. Thus as a permutation it has cycle structure

$$(0)(1)(u \ u+1).$$

The irreducible polynomial $x^2 + x + 1$ contributes exactly the length-2 cycle.

Now contrast with $x^3 + x + 1$ over $\mathbb{F}_2$. Its three roots in $\mathbb{F}_8$ form one Frobenius orbit of length 3. The difference between an irreducible quadratic and irreducible cubic is literally visible as the difference between a 2-cycle and a 3-cycle.



Deep intuition

The phrase “Frobenius encodes factorization” can sound abstract until you see this. Irreducible factors are simply orbit packets. Their degrees are the number of Frobenius steps needed to come home.

11.6 Laboratory VI: the same structural move in several subjects

A surprisingly small number of patterns recur throughout the whole refresher.

Pattern	Concrete appearance	Deep idea
Forget information	$G \rightarrow G/N, R \rightarrow R/I$	Declare a controlled family of differences invisible; kernels record exactly what was forgotten.
Add missing solutions	$K \subset K(\alpha)$	Enlarge the scalar vocabulary just enough to make an equation solvable.
Complete a configuration	splitting field	One root is not enough for symmetry; include every conjugate root.
Turn structure into motion	group action, Galois action	Abstract elements become permutations of concrete objects.
Detect convergence internally	Cauchy condition	Compare late approximations with one another rather than with an unknown limit.
Control all points at once	uniform convergence	Replace many point-dependent deadlines by one global deadline.
Turn boundary into interior	Cauchy formula	Holomorphicity converts global boundary data into local values and derivatives.
Turn products into sums	logarithmic derivative	Multiplicative arithmetic becomes additive analytic data.
Remove coordinate choices	conjugacy class	What changes by conjugation is representation-dependent; the class is intrinsic.

When a new theorem later feels unfamiliar, ask which of these old structural moves it is using. Recognizing the move is often more valuable than recognizing the theorem name.

Part VII

**Practice: reconstructing rather than
recognizing**

Conceptual mastery problems

The problems below are chosen to expose definitions and proof mechanisms rather than merely test computation. Try each without notes. Before writing details, write one sentence naming the decisive idea.

12.1 Proof language

Understanding check 12.1. Negate the statement

$$\forall x \in X \exists y \in Y \forall z \in Z : P(x, y, z).$$

Then explain the negation in plain language.

Understanding check 12.2. Give a counterexample to “every continuous function on an interval attains a maximum.” Repair the theorem by adding the correct standard hypothesis, and explain why that hypothesis addresses the failure in your counterexample.

12.2 Groups and rings

Understanding check 12.3. Prove every subgroup of index 2 is normal. Give two different conceptual explanations: one using left and right cosets, and one using the action on the two cosets.

Understanding check 12.4. Let $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/12\mathbb{Z}$ be reduction modulo 12. Identify kernel and image, then explain the first isomorphism theorem in one sentence as a statement about information loss.

Understanding check 12.5. Determine the conjugacy classes of S_3 and compute the stabilizer of each element under conjugation. Verify orbit–stabilizer for each class.

Understanding check 12.6. Prove that $\mathbb{Z}[x]/(x, 6) \cong \mathbb{Z}/6\mathbb{Z}$. Do not simply quote a standard quotient identity; define an explicit homomorphism and compute its kernel.

Understanding check 12.7. In $R = \mathbb{Z}$, determine which ideals (n) are prime and which are maximal. Then contrast with the ideal (0) .

12.3 Fields and Galois theory

Understanding check 12.8. Compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$ from a tower and justify why $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$.

Understanding check 12.9. Find the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$, and prove minimality rather than merely finding a polynomial that vanishes.

Understanding check 12.10. Show explicitly that

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}).$$

Identify the map, kernel, image, and the element of the quotient corresponding to $\sqrt{2}$.

Understanding check 12.11. Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Write all four $\mathbb{Q}$ -automorphisms and identify the fixed field of each order-two subgroup.

Understanding check 12.12. Construct $\mathbb{F}_4$ as a quotient of $\mathbb{F}_2[x]$. Write its full addition-relevant relation, compute the Frobenius permutation, and explain why its nontrivial orbit has length 2.

12.4 Real and complex analysis

Understanding check 12.13. For $f_n(x) = x^n$ on $[0, 1]$, find the pointwise limit, prove convergence is not uniform, and prove uniform convergence on every $[0, a]$ with $a < 1$. Identify the exact quantifier that fails on $[0, 1]$.

Understanding check 12.14. Give an example of differentiable functions converging uniformly whose derivatives do not converge to the derivative of the limit. Explain intuitively why differentiation amplifies the defect.

Understanding check 12.15. Find the radius of convergence and boundary behavior of

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} z^n.$$

Understanding check 12.16. Evaluate

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{e^z}{z-1} dz$$

and state exactly why Cauchy's integral formula applies.

Understanding check 12.17. Compute

$$\int_{|z|=3} \frac{z^2 + 1}{z(z-1)(z-2)} dz$$

by residues. Then recompute it by partial fractions and elementary circle integrals as a consistency check.

Understanding check 12.18. If f has a zero of order m at a , prove directly from a factorization $f = (z - a)^m g$ that f'/f has residue m . Explain why this makes f'/f a zero-counting device.

12.5 Synthesis

Understanding check 12.19. Over $\mathbb{F}_2$, prove $x^3 + x + 1$ is irreducible. If α is a root, list its Frobenius conjugates and explain why the orbit returns after exactly three steps.

Understanding check 12.20. For s with $\Re(s) > 1$, expand the finite Euler product

$$(1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1}$$

and characterize exactly which integers occur.

Understanding check 12.21. Factor $x^3 - 2$ modulo 5, 7, and 31, and match the factorization types to conjugacy classes in S_3 . Explain why this is a prototype for Chebotarev rather than the theorem itself.

Solutions and proof reconstructions

13.1 Proof language solutions

Problem 1. The negation is

$$\exists x \in X \forall y \in Y \exists z \in Z : \neg P(x, y, z).$$

Plainly: there is some x for which no matter which y is chosen, one can find a z that makes P fail. Notice how every quantifier flips.

Problem 2. On the open interval $(0, 1)$, the continuous function $f(x) = x$ has no maximum: values can approach 1 without attaining it. The repaired theorem is the extreme value theorem: a continuous real-valued function on a *compact* interval $[a, b]$ attains both maximum and minimum. Compactness repairs the missing endpoint/escape phenomenon by ensuring convergent subsequences have limits that remain inside the domain.

13.2 Group and ring solutions

Problem 3. If $H \leq G$ has index 2, there are exactly two left cosets: H and $G \setminus H$. There are also exactly two right cosets, again H and its complement. Therefore for any $g \notin H$, both gH and Hg equal $G \setminus H$, while for $g \in H$ both equal H . Hence $gH = Hg$ for all g and $H \triangleleft G$.

For the action viewpoint, G acts on the two left cosets, giving a homomorphism $G \rightarrow S_2$. The stabilizer of the coset H is exactly H . Since S_2 is abelian, the kernel has index at most 2; more directly the action is transitive and H is the kernel when the action is nontrivial, hence normal.

Problem 4. For $\varphi(k) = k \bmod 12$, the kernel is $12\mathbb{Z}$ and the image is all of $\mathbb{Z}/12\mathbb{Z}$. Thus

$$\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/\ker \varphi \cong \text{im } \varphi.$$

Conceptually, reduction modulo 12 forgets exactly which multiple of 12 was present; after that forgotten information is collapsed, the quotient is exactly the visible output.

Problem 5. Under conjugation, the classes of S_3 are

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

The stabilizer is the centralizer. For e it is all S_3 , size 6, and the orbit has size 1. For a transposition, say (12) , the centralizer is $\{e, (12)\}$, size 2, so the orbit has size $6/2 = 3$. For (123) , the centralizer is $\langle (123) \rangle$, size 3, so the orbit has size $6/3 = 2$.

Problem 6. Define

$$\Phi : \mathbb{Z}[x] \rightarrow \mathbb{Z}/6\mathbb{Z}, \quad f(x) \mapsto f(0) \bmod 6.$$

It is surjective. Its kernel consists of polynomials whose constant term is divisible by 6, precisely the ideal $(x, 6)$. Hence the first isomorphism theorem gives

$$\mathbb{Z}[x]/(x, 6) \cong \mathbb{Z}/6\mathbb{Z}.$$

Problem 7. For $n \geq 2$, (n) is prime exactly when $\mathbb{Z}/(n)$ is an integral domain, which occurs exactly when n is prime. It is maximal exactly when $\mathbb{Z}/(n)$ is a field, again exactly when n is prime. The ideal (0) is prime because $\mathbb{Z}$ is a domain, but not maximal because $\mathbb{Z}$ is not a field.

13.3 Field and Galois solutions

Problem 8. The tower

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$$

has first degree 2. If $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$, then squaring gives

$$3 = a^2 + 2b^2 + 2ab\sqrt{2}.$$

Thus $ab = 0$. If $b = 0$, then $a^2 = 3$, impossible in $\mathbb{Q}$; if $a = 0$, then $2b^2 = 3$, also impossible in $\mathbb{Q}$. Hence $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ and its minimal polynomial over that field still has degree 2. The tower law gives degree 4.

Problem 9. For $\alpha = \sqrt{2} + \sqrt{3}$,

$$\alpha^4 - 10\alpha^2 + 1 = 0.$$

From

$$\sqrt{2} = \frac{\alpha^2 - 1}{2\alpha}, \quad \sqrt{3} = \frac{\alpha^2 + 1}{2\alpha},$$

we get $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which has degree 4 by Problem 8. Therefore the minimal polynomial has degree 4, so it is

$$x^4 - 10x^2 + 1.$$

Problem 10. Evaluation at $\sqrt{2}$ gives

$$\text{ev}_{\sqrt{2}} : \mathbb{Q}[x] \rightarrow \mathbb{Q}(\sqrt{2}), \quad f \mapsto f(\sqrt{2}).$$

Its kernel is $(x^2 - 2)$ because $x^2 - 2$ is the minimal polynomial. It is surjective because every $a + b\sqrt{2}$ is the image of $a + bx$. Therefore

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}),$$

and the class of x corresponds to $\sqrt{2}$.

Problem 11. The four automorphisms independently choose signs:

$$(\sqrt{2}, \sqrt{3}) \mapsto (\pm\sqrt{2}, \pm\sqrt{3}).$$

The subgroup flipping only $\sqrt{2}$ fixes $\mathbb{Q}(\sqrt{3})$; the subgroup flipping only $\sqrt{3}$ fixes $\mathbb{Q}(\sqrt{2})$; the subgroup flipping both fixes $\mathbb{Q}(\sqrt{6})$. Degree considerations show these are the full fixed fields.

Problem 12. Take

$$\mathbb{F}_4 = \mathbb{F}_2[u]/(u^2 + u + 1),$$

where $u^2 = u + 1$. The elements are $0, 1, u, u + 1$. Frobenius is squaring and sends

$$u \mapsto u^2 = u + 1, \quad u + 1 \mapsto u^2 + 1 = u.$$

Thus its nontrivial orbit is $\{u, u + 1\}$ of length 2, matching the degree of the irreducible polynomial that creates u .

13.4 Analysis solutions

Problem 13. The pointwise limit of x^n on $[0, 1]$ is 0 for $x < 1$ and 1 at $x = 1$. It is not uniform because with $x_n = 2^{-1/n}$ one has $x_n < 1$ but $x_n^n = 1/2$, so the uniform error never falls below $1/2$. On $[0, a]$, $a < 1$,

$$\sup_{x \in [0, a]} x^n = a^n \rightarrow 0.$$

The failed quantifier is the demand for a single N that works for every x simultaneously.

Problem 14. The functions $f_n(x) = \sin(nx)/n$ satisfy $\|f_n\|_\infty \leq 1/n$, so $f_n \rightarrow 0$ uniformly, but $f'_n(x) = \cos(nx)$ does not converge appropriately. Differentiation measures local change and multiplies the frequency n back into the function, undoing the small amplitude factor.

Problem 15. For $a_n = n^2/3^n$,

$$\limsup |a_n|^{1/n} = \frac{1}{3},$$

so the radius is 3. On $|z| = 3$, the term magnitude is n^2 , so the terms do not tend to zero; the series diverges at every boundary point.

Problem 16. The circle $|z| = 2$ contains 1, and e^z is holomorphic on a neighborhood of the closed disk. Therefore Cauchy's formula gives

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{e^z}{z-1} dz = e.$$

Problem 17. The residues at 0, 1, 2 are respectively $1/2, -2, 5/2$, summing to 1. Hence the integral is $2\pi i$. Partial fractions has the form

$$\frac{z^2 + 1}{z(z-1)(z-2)} = \frac{1/2}{z} - \frac{2}{z-1} + \frac{5/2}{z-2},$$

and each elementary circle integral is $2\pi i$ times its coefficient, yielding the same answer.

Problem 18. Write $f = (z-a)^m g$ with $g(a) \neq 0$. Then

$$\frac{f'}{f} = \frac{m}{z-a} + \frac{g'}{g}.$$

The second term is holomorphic near a , so the residue is m . A contour integral of f'/f therefore adds zero multiplicities inside the contour, which is the local mechanism behind the argument principle.

13.5 Synthesis solutions

Problem 19. At 0 and 1, the polynomial $x^3 + x + 1$ has value 1, so it has no root in $\mathbb{F}_2$. A reducible cubic over a field must have a linear factor, hence a root; therefore it is irreducible. If α is a root in $\mathbb{F}_8$, Frobenius gives

$$\alpha, \alpha^2, \alpha^4,$$

and the next iterate is $\alpha^8 = \alpha$. No earlier return is possible because the minimal polynomial has degree 3.

Problem 20. Expand the geometric series:

$$(1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1} = \sum_{a,b \geq 0} 2^{-as} 3^{-bs} = \sum_{a,b \geq 0} (2^a 3^b)^{-s}.$$

Exactly the positive integers with no prime factors other than 2 and 3 occur, each once by unique factorization.

Problem 21. Modulo 5,

$$x^3 - 2 = (x+2)(x^2 - 2x - 1),$$

with irreducible quadratic, so the type is 1 + 2, matching a transposition. Modulo 7 the cubic has no root, hence is irreducible, matching a 3-cycle. Modulo 31,

$$x^3 - 2 = (x-4)(x-7)(x+11),$$

matching the identity. This is a prototype because we are seeing the finite-field factorization/cycle dictionary, but Chebotarev is the global density theorem asserting how often each conjugacy class occurs for unramified primes.

A compact recall deck

Use these prompts closed-book. A definition is not secure until you can state it, motivate it, give an example and a nonexample, and name a theorem that uses it.

Prompt	What you should be able to reconstruct
Quantifier negation	Flip every quantifier and negate the terminal predicate. Explain the game-of-choices interpretation.
Contrapositive	$P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$; the converse is generally different.
Subgroup	Nonempty and closed under ab^{-1} . Explain why this recovers identity, inverse, and product.
Homomorphism	Operation-preserving map. Kernel = lost information; image = surviving information.
Normal subgroup	$gNg^{-1} = N$. Explain why this is exactly quotient well-definedness.
Orbit/stabilizer	Destinations versus symmetries fixing the point; $ G = \text{Orb}(x) \text{Stab}(x) $.
Ideal	Additive subgroup absorbing multiplication from all of R ; precisely what quotient multiplication needs.
Prime ideal	Quotient is a domain.
Maximal ideal	Quotient is a field.
CRT	Comaximal ideals give independent residue coordinates.
Field extension	Larger field as vector space over base; degree is dimension.
Minimal polynomial	Shortest monic algebraic identity; degree equals simple extension degree.
Splitting field	Smallest field containing the complete root configuration.
Separable	Conjugate roots are distinct.
Normal	If one conjugate lies in the extension, all do.
Galois	Finite + separable + normal; automorphisms expose root symmetry.
Fixed field	Elements invisible to a subgroup of automorphisms.
Frobenius	$x \mapsto x^q$; cycles through finite-field conjugates.
Sequence limit	Every tolerance eventually contains the whole tail.
Cauchy	Late terms become mutually close; convergence detected without knowing the limit.
Uniform convergence	One accuracy deadline works for every point simultaneously.
M-test	Dominates functions by a convergent numerical envelope.

Power-series radius	Root test applied to coefficients; compactly inside the disk gives uniform convergence.
Holomorphic	One complex derivative must work from every direction; locally a rotation-scaling.
Cauchy formula	Boundary values determine interior values and all derivatives.
Laurent series	Taylor series adapted to a punctured neighborhood.
Residue	Coefficient of $(z - a)^{-1}$, the unique Laurent mode detected by a loop.
Logarithmic derivative	Converts products into sums; residues count zeros and poles.
Euler product	Unique factorization reorganizes integers prime-by-prime.
Chebotarev preview	prime $\rightarrow$ residue field $\rightarrow$ Frobenius $\rightarrow$ conjugacy class $\rightarrow$ analytic density.

A ten-session refresher plan

This plan is a default, not a deadline. Each session should end with closed-book reconstruction rather than rereading.

Day	Focus	Required reconstruction
1	Proof language	Negate nested quantifiers; prove one direct, contrapositive, contradiction, and induction example.
2	Groups and quotients	Rebuild the subgroup test, first isomorphism theorem, and quotient well-definedness from definitions.
3	Actions and Sylow	Compute S_3 conjugacy classes; derive orbit-stabilizer; explain both constraints on n_p .
4	Rings and CRT	Explain ideals as kernels/erasable quantities; prove prime/maximal quotient criteria and CRT.
5	Fields	Derive the tower law and quotient construction; compute two minimal polynomials.
6	Galois/Frobenius	Reconstruct $x^3 - 2$, the $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ correspondence, $\mathbb{F}_4$, and Frobenius cycles.
7	Real convergence	State limit/Cauchy quantifiers; prove completeness route; classify several series.
8	Uniform convergence	Explain the moving-spike obstruction; prove continuity/integration preservation and M-test; derive power-series radius.
9	Complex analysis	Reconstruct CR equations, Goursat proof idea, Cauchy formula, Cauchy estimates, and Liouville.
10	Residues + synthesis	Derive residue theorem from principal parts; explain logarithmic derivative; factor $x^3 - 2$ modulo 5, 7, 31 and connect to Frobenius classes.

Correctness audit notes

This revision deliberately checks several places where refresher texts often hide a hypothesis or create a circular proof. The following checkpoints summarize the choices made in the main text.

- **Quotient groups.** Normality is proved both sufficient and necessary for representative-independent coset multiplication; it is not presented merely as a convenient assumption.
- **Sylow theory.** The existence proof tracks the p -part of the stabilizer carefully, and the congruence $n_p \equiv 1 \pmod{p}$ uses the conjugation action with the unique fixed Sylow subgroup explicitly identified.
- **Minimal polynomials.** Concrete calculations distinguish an annihilating polynomial from a minimal polynomial by proving the extension degree or irreducibility.
- **Finite fields.** Existence and uniqueness of $\mathbb{F}_q$ are proved from the splitting field of $x^q - x$. For $\mathbb{F}_{q^n}/\mathbb{F}_q$, the degree n , separability, normality, and exact order of Frobenius are all justified before identifying the Galois group.
- **Galois correspondence.** The degree equalities are derived from the embedding count and Artin fixed-field theorem rather than assumed as part of the correspondence.
- **Uniform convergence.** Continuity and integration use uniform convergence directly; differentiation is given its stronger, correct hypothesis: uniform convergence of the derivatives plus convergence at one anchor point.
- **Identity theorem.** The local zero dichotomy is proved first, then the identity theorem, and only afterward the global statement that zeros of a nonzero holomorphic function are isolated. This avoids the common circular argument in which isolated zeros are used to prove the identity theorem while the isolation claim itself depends on it.
- **Cauchy theorem and topology.** The text proves star-shaped Cauchy theory directly. The later homotopy statement uses an explicitly piecewise- C^2 notion of homotopy, so no unproved equivalence with a more general topological definition of simple connectivity is silently required.
- **Laurent and residue theory.** The annular representation makes the inner and outer boundary contributions explicit. The residue theorem on a circle is then proved by subtracting full principal parts, reducing the remainder to the already proved disk version of Cauchy's theorem.
- **Euler products.** The product is defined as the limit of finite prime products in $\Re(s) > 1$. The logarithmic derivative is justified through a locally uniformly convergent double series and an exponential identity, avoiding an implicit global choice of a logarithm branch.

These checks do not make the book foundational in the set-theoretic sense: the completeness axiom for $\mathbb{R}$ and standard finite-dimensional linear algebra remain declared background. They do make the logical dependencies of the material taught here explicit.

Final perspective

The central lesson is not that Chebotarev requires a long list of prerequisites. It is that the theorem lies at a meeting point of a small number of structural ideas:

- **quotients** formalize what information we choose to forget;
- **field extensions** formalize what algebraic information we add;
- **Galois groups** record symmetries of the added root structure;
- **finite-field Frobenius** turns factorization into a canonical symmetry;
- **conjugacy** removes dependence on auxiliary choices;
- **uniform convergence and holomorphicity** justify analytic manipulation;
- **Euler products and logarithmic derivatives** translate prime factorization into analyzable functions.

If these ideas feel connected rather than memorized, the reboot has done its job.